Local, Boundary, and Reciprocal Expansions of q-Pochhammer Symbols, Gaussian Coefficients, and Related q-Analogues

Expansions at $q = 1$, $q = 0$, roots of unity, and $q = \infty$

Research report for the ProveIt Fabius-function project

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Abstract

This report develops a unified calculus for expansions of finite and infinite q-products and the q-analogues built from them. Four qualitatively different situations are separated from the outset: ordinary Taylor expansions of finite q-polynomials; boundary asymptotics of infinite products as q approaches the unit circle; Laurent expansions at the point $q = \infty$ for finite objects; and generalized power or transseries expansions when a parameter coalesces with a zero of the product. This distinction prevents several common but consequential confusions, such as treating $(q; q)_\infty$ as analytic at $q = 1$, or speaking of an unrenormalized infinite product at $|q| > 1$.

For finite shifted factorials we give all-order expansions both with a fixed parameter a and in the classical scaling $a = q^\alpha$. For Gaussian and multinomial coefficients, the logarithmic coefficients are expressed by Bernoulli numbers and Faulhaber differences, while complete exponential Bell polynomials recover every ordinary coefficient. The centered Gaussian coefficient is shown to be an even function of $t = -\log q$, and its coefficients are identified with exact cumulants of the inversion statistic. Around $q = 0$ the coefficients are weighted restricted-partition numbers; reciprocity then transfers the entire expansion to $q = \infty$ and distinguishes the positive and negative real rays. At $q = -1$ we obtain not only the familiar value of a Gaussian coefficient but its complete first jet, including a closed derivative formula in the vanishing parity case. At a general root of unity, cyclotomic factorization gives the zero order and a Bell-polynomial algorithm for all Taylor coefficients.

For infinite products, the fixed-parameter $q \to 1$ expansion is written in polylogarithms, and the coalescing specialization $a = q^x$ is derived by Mellin inversion. This yields explicit all-order expansions of the q-gamma and q-beta functions. Dedekind-eta modularity supplies exponentially improved formulae at $q = 1$ and $q = -1$. A radial expansion at every primitive root of unity is derived directly from the Lambert series; its constant term is a cyclic quantum dilogarithm. Further sections treat the two standard q-exponentials, the q-difference operator, basic hypergeometric series as differential-operator deformations of ordinary hypergeometric series, and a uniform double-scaling regime $q = \exp(-\tau/N),\ k \sim \alpha N$ for Gaussian and multinomial coefficients. The latter has an all-order expansion in odd powers of N^{-1} whose coefficients are elementary polylogarithmic expressions.

The report was prepared as a self-contained extension of the asymptotic chapter of the ProveIt monograph on q-Pochhammer symbols and q-binomial coefficients [1]. A companion Python program performs exact symbolic checks and high-precision numerical tests of the principal new formulae.

Principal results and status

The following guide distinguishes identities proved in the report from asymptotic statements and from proposed research directions.

Topic	Main output	Status
Finite q-products at regular points	Bell-polynomial Taylor formula from logarithmic derivatives; explicit treatment of resonant zeros	Exact
$q \to 1$, finite products	Fixed-a polylogarithmic coefficients and scaled $a = q^\alpha$ Bernoulli-polynomial coefficients	Exact locally / convergent
Gaussian coefficients	All logarithmic coefficients, centered even expansion, ordinary $(q-1)$ Taylor coefficients, cumulants	Exact
$q \to 0$	Weighted distinct-partition and restricted-partition expansions	Exact
$q \to \infty$	Complete reciprocal Laurent expansions; separate signs on the two real rays	Exact after normalization
$q = -1$	Closed value and first derivative for every n, k	Exact
General roots of unity	Cyclotomic order, carry criterion, and all Taylor jets	Exact for finite objects
Infinite product near $q = 1$	Polylogarithmic expansion for fixed a; Mellin expansion for $a = q^x$	Asymptotic
q-gamma and q-beta	Explicit all-order logarithmic expansions	Asymptotic
Root-of-unity radial limit	Cyclic-dilogarithm leading factor and all coefficient sums	Asymptotic
Double scaling	All-order expansion for $q = \mathrm{e}^{-\tau/N}$ and $k = \alpha N$	Asymptotic, uniform on compact parameter sets
Research programme	Uniform resonant scaling, higher cyclotomic jets, resurgence, formal verification	Conjectural / proposed

Contents

Chapter 1

What kind of expansion is meant?

1.1 Four analytically different regimes

The phrase "expand at $q = q_0$" hides several inequivalent operations. Throughout the report they are kept separate.

(i) A finite shifted factorial $(a;q)_n$ and a Gaussian coefficient $\begin{bmatrix} n \\ k \end{bmatrix}_q$ are polynomials in q. They therefore possess ordinary, globally convergent Taylor series at every finite q_0. The logarithm of such a polynomial is only local and is unavailable at a zero.

(ii) An infinite product $(a;q)_\infty$ is naturally defined for $|q| < 1$. The points $q = 1$ and the other points of the unit circle are boundary points, not ordinary Taylor centers. The appropriate objects are sectorial asymptotic expansions, often with exponentials such as $\exp(-\operatorname{Li}_2(a)/t)$ and, in resonant cases, powers and logarithms of $t = -\log q$.

(iii) A finite polynomial has a Laurent expansion at the point $q = \infty$ on the Riemann sphere. Reciprocity makes this expansion a reversed copy of its Taylor series at $q = 0$. The symbols $+\infty$ and $-\infty$ describe two real approaches to the same complex point and differ only through the phases of the inverse powers.

(iv) When a parameter depends on q and approaches a zero of a product, the expansion generally leaves the category of ordinary powers. Examples are $a = q^x$ as $q \to 1$ and $\Gamma_q(x)$ as $q \to 0$ for nonintegral x. Powers q^x, logarithms, and exponentially small corrections then form the natural asymptotic scale.

This taxonomy is not merely terminological. It determines which algebraic operations are legitimate, what the remainder means, and whether an expansion can be continued through the proposed center.

1.2 Notation

For $n \in \mathbb{N}_0$,

$$(a;q)_n = \prod_{j=0}^{n-1}(1 - aq^j), \qquad (a;q)_\infty = \prod_{j=0}^{\infty}(1 - aq^j) \quad (|q| < 1),$$

and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_k (q;q)_{n-k}} \qquad (0 \le k \le n),$$

where the quotient is understood as its polynomial continuation. We write

$$S_r(N) = \sum_{j=1}^{N} j^r, \qquad \Delta_r(n,k) = S_r(n) - S_r(k) - S_r(n-k), \qquad D = k(n-k).$$

The Bernoulli numbers and polynomials use the convention

$$\frac{z\mathrm{e}^{xz}}{\mathrm{e}^z - 1} = \sum_{r=0}^{\infty} B_r(x)\frac{z^r}{r!}, \qquad B_r = B_r(0),$$

so $B_1 = -1/2$. The polylogarithm is $\mathrm{Li}_s(z) = \sum_{m\geq 1} z^m/m^s$ in its initial disk and by analytic continuation elsewhere.

Two local coordinates will recur:

$$q = \mathrm{e}^{-t}, \qquad t = -\log q, \qquad \text{and} \qquad h = q - 1.$$

The logarithmic coordinate t exposes reciprocity and Bernoulli structure; h gives the literal Taylor expansion in q.

1.3 Bell polynomials and conversion of coordinates

Definition 1.1 (Complete exponential Bell polynomials)**.** The polynomials $\mathcal{Y}_r(x_1, \ldots, x_r)$ are defined by

$$\exp\left(\sum_{j\geq 1} x_j \frac{z^j}{j!}\right) = \sum_{r\geq 0} \mathcal{Y}_r(x_1, \ldots, x_r)\frac{z^r}{r!}.$$

Thus $\mathcal{Y}_0 = 1$, $\mathcal{Y}_1 = x_1$, $\mathcal{Y}_2 = x_1^2 + x_2$, and $\mathcal{Y}_3 = x_1^3 + 3x_1x_2 + x_3$.

Lemma 1.2 (Exponentiation of a logarithmic expansion)**.** *If*

$$\log\frac{F(z)}{F(0)} = \sum_{j\geq 1} \kappa_j \frac{z^j}{j!}, \qquad F(0) \neq 0,$$

then

$$F(z) = F(0)\sum_{r\geq 0} \mathcal{Y}_r(\kappa_1, \ldots, \kappa_r)\frac{z^r}{r!}.$$

The identity is formal; whenever the logarithmic series converges, the resulting series converges in the same local disk.

Proof. This is the defining generating function of the complete exponential Bell polynomials after substituting $x_j = \kappa_j$. □

Let $\vartheta = q\frac{\mathrm{d}}{\mathrm{d}q}$. Since $q = \mathrm{e}^{-t}$,

$$\frac{\mathrm{d}}{\mathrm{d}t}F(\mathrm{e}^{-t}) = -\vartheta F(q).$$

Ordinary derivatives at $q = 1$ can be recovered from logarithmic-coordinate derivatives by

$$F^{(r)}(1) = \sum_{j=0}^{r} s(r,j)(-1)^j \left.\frac{\mathrm{d}^j}{\mathrm{d}t^j}F(\mathrm{e}^{-t})\right|_{t=0}, \tag{1.1}$$

where $s(r,j)$ are the signed Stirling numbers of the first kind. Indeed, $q^r\frac{\mathrm{d}^r}{\mathrm{d}q^r} = \vartheta(\vartheta-1)\cdots(\vartheta-r+1)$. Formula (1.1), together with Theorem 1.2, is an all-order converter between a compact cumulant-like expansion in t and a literal Taylor series in $q-1$.

Chapter 2

Finite products at an arbitrary center

2.1 The universal regular-point formula

Theorem 2.1 (Taylor expansion from logarithmic derivatives)**.** *Let $P(q)$ be holomorphic near q_0 and suppose $P(q_0) \neq 0$. Put*

$$\lambda_r(q_0) = \frac{\mathrm{d}^r}{\mathrm{d}q^r} \log P(q)\bigg|_{q=q_0}.$$

Then

$$P(q_0 + h) = P(q_0) \sum_{r \geq 0} \frac{\mathcal{Y}_r(\lambda_1, \ldots, \lambda_r)}{r!} h^r.$$

For a polynomial the Taylor series terminates, although the logarithmic series itself generally does not.

Proof. Apply Theorem 1.2 to $F(h) = P(q_0 + h)$. □

For the finite shifted factorial, logarithmic differentiation may be performed factor by factor:

$$\frac{\mathrm{d}}{\mathrm{d}q} \log(a;q)_n = -\sum_{j=1}^{n-1} \frac{ajq^{j-1}}{1 - aq^j}. \tag{2.1}$$

Higher derivatives are rational functions with possible poles only at zeros of the product. A useful diagonal form follows from $\vartheta = q\,\mathrm{d}/\,\mathrm{d}q$:

$$\vartheta^r \log(a;q)_n = -\sum_{m \geq 1} a^m m^{r-1} \sum_{j=0}^{n-1} j^r q^{jm} \qquad (r \geq 1), \tag{2.2}$$

initially when every $|aq^j| < 1$ and elsewhere by analytic continuation. The finite geometric power sum in (2.2) can be reduced to a rational function of q^m or to Eulerian polynomials.

2.2 Resonant zeros of a finite shifted factorial

At a zero, the logarithmic expansion must be replaced by a factored expansion. The zero multiplicity is completely elementary but becomes nontrivial at roots of unity, where several factors may vanish simultaneously.

Theorem 2.2 (Local leading term at a resonant point). *Fix $n \geq 1$, $q_0 \neq 0$, and $a \neq 1$. Let*

$$J = \{j \in \{1, \ldots, n-1\} : aq_0^j = 1\}, \qquad r = |J|.$$

Then

$$(a;q)_n = C(a, q_0, n)(q - q_0)^r \big(1 + \mathcal{O}(q - q_0)\big),$$

where

$$C(a, q_0, n) = (-1)^r q_0^{-r} \left(\prod_{j \in J} j\right) \left(\prod_{\substack{0 \leq j < n \\ j \notin J}} (1 - aq_0^j)\right). \tag{2.3}$$

After extracting the displayed power, all further coefficients follow from Theorem 2.1 applied to the nonvanishing quotient.

Proof. For $j \in J$,

$$1 - aq^j = -ajq_0^{j-1}(q - q_0) + \mathcal{O}((q - q_0)^2) = -\frac{j}{q_0}(q - q_0) + \mathcal{O}((q - q_0)^2).$$

Multiplying these r linear factors and evaluating every other factor at q_0 gives (2.3). The quotient by $(q - q_0)^r$ is holomorphic and nonzero at q_0, so the regular-point theorem applies. $\square$

Remark 2.3. If $a = 1$, the factor with $j = 0$ vanishes identically and $(1;q)_n = 0$ for all q and all $n \geq 1$. The important object $(q;q)_n$ is different: its first parameter varies with q and is treated by the scaled analysis in Chapter 3.

2.3 Gaussian coefficients through cyclotomic factors

The polynomial factorization

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{d=1}^{n} \Phi_d(q)^{e_d(n,k)}, \qquad e_d(n,k) = \left\lfloor \frac{n}{d} \right\rfloor - \left\lfloor \frac{k}{d} \right\rfloor - \left\lfloor \frac{n-k}{d} \right\rfloor, \tag{2.4}$$

has $e_d(n,k) \in \{0, 1\}$. It is therefore an optimal local representation at roots of unity.

Proposition 2.4 (Regular Taylor jets of a Gaussian polynomial). *If $G_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q$ and $G_{n,k}(q_0) \neq 0$, then*

$$(\log G_{n,k})^{(r)}(q_0) = \sum_{d : e_d(n,k) = 1} (\log \Phi_d)^{(r)}(q_0).$$

Consequently every Taylor coefficient at q_0 is an explicit Bell polynomial in logarithmic derivatives of the selected cyclotomic polynomials.

Proof. Take a local logarithm of (2.4) and apply Theorem 2.1. $\square$

This construction is denominator-free and remains meaningful when the usual factorial quotient is of the indeterminate form $0/0$.

Chapter 3

Finite products as $q \to 1$

3.1 A fixed first parameter

The source monograph emphasizes the scaled factors that build Gaussian coefficients. A complementary expansion keeps a fixed while q moves. It is useful for perturbing terminating basic hypergeometric identities whose parameters are not integral powers of q.

Theorem 3.1 (Fixed-a finite shifted factorial)**.** *Let $n \geq 1$, $q = \mathrm{e}^{-t}$, and initially assume $|a| < 1$. Then, near $t = 0$,*

$$\log(a;q)_n = n\log(1-a) + \sum_{r\geq 1}(-1)^{r+1}S_r(n-1)\,\mathrm{Li}_{1-r}(a)\frac{t^r}{r!}. \tag{3.1}$$

The identity extends analytically in a to every local domain avoiding the zeros of $(a;q)_n$ at $q = 1$. The ordinary coefficients of $(a;q)_n$ are the Bell polynomials in

$$\kappa_r = (-1)^{r+1}S_r(n-1)\,\mathrm{Li}_{1-r}(a).$$

Proof. For $|a| < 1$,

$$\log(1-a\mathrm{e}^{-jt}) = -\sum_{m\geq 1}\frac{a^m}{m}\mathrm{e}^{-jmt}.$$

Expanding the exponential and summing over $0 \leq j < n$ gives

$$\begin{aligned}\log(a;q)_n &= -\sum_{m\geq 1}\frac{a^m}{m}\sum_{j=0}^{n-1}\sum_{r\geq 0}\frac{(-jmt)^r}{r!}\\ &= n\log(1-a) + \sum_{r\geq 1}(-1)^{r+1}\left(\sum_{j=0}^{n-1}j^r\right)\left(\sum_{m\geq 1}a^m m^{r-1}\right)\frac{t^r}{r!},\end{aligned}$$

which is (3.1). The sum is finite in j, so the result is a local analytic identity. Analytic continuation in a and Theorem 1.2 finish the proof. ☐

The first terms are

$$\begin{aligned}\log\frac{(a;q)_n}{(1-a)^n} &= \frac{n(n-1)}{2}\frac{a}{1-a}t - \frac{n(n-1)(2n-1)}{12}\frac{a}{(1-a)^2}t^2\\ &\quad + \frac{n^2(n-1)^2}{24}\frac{a(1+a)}{(1-a)^3}t^3 + \mathcal{O}(t^4).\end{aligned} \tag{3.2}$$

These terms already show why the fixed-a and $a = q^\alpha$ limits must not be interchanged: the rational functions of a in (3.2) become singular as $a \to 1$.

3.2 The classical scaling $a = q^\alpha$

Theorem 3.2 (Finite scaled shifted factorial). *Let $q = \mathrm{e}^{-t}$ and suppose none of $\alpha, \alpha+1, \dots, \alpha+n-1$ is zero. Define*

$$A_r(\alpha, n) = \sum_{j=0}^{n-1} (\alpha+j)^r = \frac{B_{r+1}(\alpha+n) - B_{r+1}(\alpha)}{r+1}.$$

Then

$$(q^\alpha; q)_n = t^n (\alpha)_n \exp\left(-\frac{A_1(\alpha, n)}{2} t + \sum_{r \ge 1} \frac{B_{2r} A_{2r}(\alpha, n)}{2r(2r)!} t^{2r} \right). \tag{3.3}$$

For fixed α, n the logarithmic series converges in every disk satisfying $|(\alpha+j)t| < 2\pi$ for $0 \le j < n$. As a formal expansion it remains valid after algebraic continuation, provided resonant zero factors are first extracted.

Proof. The elementary identity

$$1 - \mathrm{e}^{-x} = x \exp\left(-\frac{x}{2} + \sum_{r \ge 1} \frac{B_{2r}}{2r(2r)!} x^{2r} \right) \tag{3.4}$$

holds for $|x| < 2\pi$. Apply it to $x = (\alpha+j)t$ and multiply over $0 \le j < n$. The product of the linear factors is $t^n (\alpha)_n$, and the sums of powers are precisely $A_r(\alpha, n)$. $\square$

Example 3.3 (The ordinary q-factorial). Since $[n]_q! = (q; q)_n / (1-q)^n$, one obtains

$$\log \frac{[n]_{\mathrm{e}^{-t}}!}{n!} = -\frac{n(n-1)}{4} t + \sum_{r \ge 1} \frac{B_{2r}}{2r(2r)!} \big(S_{2r}(n) - n\big) t^{2r}. \tag{3.5}$$

For a single q-number,

$$[x]_{\mathrm{e}^{-t}} = x \exp\left(-\frac{x-1}{2} t + \sum_{r \ge 1} \frac{B_{2r}(x^{2r} - 1)}{2r(2r)!} t^{2r} \right). \tag{3.6}$$

The same formula is meaningful with x replaced by an operator, a fact used in Chapter 13.

Chapter 4

Gaussian and multinomial coefficients near $q = 1$

4.1 All logarithmic coefficients

The basic Bernoulli formula in the source monograph is recalled here only to make the later coefficient machinery self-contained.

Theorem 4.1 (Complete Gaussian expansion)**.** *Let $q = \mathrm{e}^{-t}$, $0 \le k \le n$, and $D = k(n-k)$. Then*

$$\log \frac{\left[{n \atop k}\right]_{\mathrm{e}^{-t}}}{\binom{n}{k}} = -\frac{D}{2}t + \sum_{r\ge1} \frac{B_{2r}\Delta_{2r}(n,k)}{2r(2r)!}t^{2r}. \tag{4.1}$$

For fixed n, k this converges when $|t| < 2\pi/n$. If

$$\kappa_1 = -\frac{D}{2}, \qquad \kappa_{2r} = \frac{B_{2r}}{2r}\Delta_{2r}(n,k), \qquad \kappa_{2r+1} = 0 \quad (r \ge 1),$$

then the literal coefficient expansion is

$$\begin{bmatrix} n \\ k \end{bmatrix}_{\mathrm{e}^{-t}} = \binom{n}{k} \sum_{j\ge0} \mathcal{Y}_j(\kappa_1, \dots, \kappa_j)\frac{t^j}{j!}. \tag{4.2}$$

Proof. Use the product

$$\begin{bmatrix} n \\ k \end{bmatrix}_{\mathrm{e}^{-t}} = \prod_{j=1}^{k} \frac{1 - \mathrm{e}^{-(n-k+j)t}}{1 - \mathrm{e}^{-jt}},$$

apply (3.4), and collect power sums. The linear difference is D and the even difference is $\Delta_{2r}(n,k)$. Bell exponentiation gives (4.2). □

4.2 Centering, explicit coefficients, and symmetry

Reciprocity gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_{\mathrm{e}^{-t}} = \mathrm{e}^{-Dt}\begin{bmatrix} n \\ k \end{bmatrix}_{\mathrm{e}^{t}}.$$

Therefore

$$C_{n,k}(t) = \mathrm{e}^{Dt/2} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{\mathrm{e}^{-t}}}{\binom{n}{k}} \tag{4.3}$$

is an even analytic function. Put

$$c_{2r} = \frac{B_{2r}\Delta_{2r}}{2r(2r)!}.$$

Then

$$\begin{aligned} C_{n,k}(t) = 1 &+ c_2 t^2 + \left(c_4 + \frac{c_2^2}{2}\right) t^4 \\ &+ \left(c_6 + c_2 c_4 + \frac{c_2^3}{6}\right) t^6 + \mathcal{O}(t^8). \end{aligned} \tag{4.4}$$

The needed Faulhaber differences factor as

$$\Delta_2 = D(n+1), \tag{4.5}$$

$$\Delta_4 = D(n+1)\big(n(n+1) - D\big), \tag{4.6}$$

$$\Delta_6 = \frac{D(n+1)}{2}\big(2D^2 - 4Dn^2 - 5Dn + 2n^4 + 4n^3 + n^2 - n\big). \tag{4.7}$$

Consequently

$$c_2 = \frac{D(n+1)}{24}, \qquad c_4 = -\frac{D(n+1)(n(n+1) - D)}{2880}, \qquad c_6 = \frac{\Delta_6}{181440}.$$

All higher Δ_{2r} are symmetric polynomials in k and $n-k$, hence polynomials in n and D.

Corollary 4.2 (Literal Taylor series in $h = q - 1$)**.** *As $h \to 0$,*

$$\begin{aligned} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1+h}}{\binom{n}{k}} &= 1 + \frac{D}{2}h + \frac{D(3D+n-5)}{24}h^2 \\ &+ \frac{D(D-2)(D+n-3)}{48}h^3 + \mathcal{O}(h^4). \end{aligned} \tag{4.8}$$

Every later coefficient is obtained mechanically from (4.1) *by $t = -\log(1+h)$ and Bell expansion, or by the Stirling conversion* (1.1)*.*

Proof. Substitute $t = -h + h^2/2 - h^3/3 + \mathcal{O}(h^4)$ into (4.1) and exponentiate. □

4.3 Cumulants of inversions

Let $X_{n,k}$ be the inversion number of a uniformly random binary word with k ones and $n-k$ zeros. Its probability generating function is

$$\mathbb{E}(q^{X_{n,k}}) = \frac{\begin{bmatrix} n \\ k \end{bmatrix}_q}{\binom{n}{k}}.$$

Putting $q = \mathrm{e}^s$ in (4.1) therefore gives all cumulants.

Corollary 4.3 (Exact inversion cumulants). *The odd centered cumulants vanish, and*

$$\begin{aligned} \operatorname{cum}_1(X_{n,k}) &= \frac{D}{2}, \\ \operatorname{cum}_{2r}(X_{n,k}) &= \frac{B_{2r}}{2r}\Delta_{2r}(n,k) \qquad (r \geq 1), \\ \operatorname{cum}_{2r+1}(X_{n,k}) &= 0 \qquad (r \geq 1). \end{aligned}$$

In particular,

$$\operatorname{Var}(X_{n,k}) = \frac{D(n+1)}{12},$$

and the fourth cumulant is

$$-\frac{D(n+1)(n(n+1)-D)}{120}.$$

The evenness in (4.3) is thus exactly the symmetry of the inversion distribution about $D/2$, while Bernoulli numbers encode its entire cumulant hierarchy.

4.4 Multinomial extension

Let $n_1 + \cdots + n_s = n$ and write

$$\begin{bmatrix} n \\ n_1, \ldots, n_s \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_{n_1} \cdots (q;q)_{n_s}}.$$

Define

$$\Delta_r(\boldsymbol{n}) = S_r(n) - \sum_{i=1}^{s} S_r(n_i), \qquad D(\boldsymbol{n}) = \Delta_1(\boldsymbol{n}) = \sum_{i<j} n_i n_j.$$

The same proof gives

$$\log \frac{\left[\begin{smallmatrix} n \\ n_1,\ldots,n_s \end{smallmatrix}\right]_q \big|_{q=\mathrm{e}^{-t}}}{\binom{n}{n_1,\ldots,n_s}} = -\frac{D(\boldsymbol{n})}{2}t + \sum_{r\geq 1} \frac{B_{2r}\Delta_{2r}(\boldsymbol{n})}{2r(2r)!}t^{2r}. \tag{4.9}$$

Thus the centered q-multinomial is even, and $B_{2r}\Delta_{2r}(\boldsymbol{n})/(2r)$ is the $2r$th cumulant of the inversion number of a uniformly random multiset permutation.

Chapter 5

Expansions at $q = 0$

5.1 Finite and infinite shifted factorials

For $n \geq 1$,

$$(a;q)_n = (1-a)\prod_{j=1}^{n-1}(1-aq^j).$$

Every coefficient therefore has a direct restricted-partition meaning.

Theorem 5.1 (Weighted distinct-partition expansion). *For $r \geq 0$ define*

$$d_{n,r}(a) = \sum_{\substack{J \subseteq \{1,\ldots,n-1\} \\ \sum_{j\in J} j = r}} (-a)^{|J|}.$$

Then

$$(a;q)_n = (1-a)\sum_{r=0}^{n(n-1)/2} d_{n,r}(a)q^r. \tag{5.1}$$

For fixed r, $d_{n,r}(a)$ stabilizes once $n > r$ to

$$d_r(a) = \sum_{\substack{\lambda \vdash r \\ \lambda \text{ has distinct parts}}} (-a)^{\ell(\lambda)},$$

and hence

$$(a;q)_\infty = (1-a)\sum_{r\geq 0} d_r(a)q^r. \tag{5.2}$$

Proof. Expand the finite product by choosing either 1 or $-aq^j$ from each factor. A choice of indices is a subset J and contributes $(-a)^{|J|}q^{\sum J}$. Factors with index at least n cannot affect a coefficient of degree below n, which proves stabilization and the infinite identity. □

The beginning of the stable expansion is

$$\begin{aligned}\frac{(a;q)_\infty}{1-a} &= 1 - aq - aq^2 + a(a-1)q^3 + a(a-1)q^4 \\ &\quad + a(2a-1)q^5 - a(a-1)^2q^6 - a(a^2-3a+1)q^7 + \mathcal{O}(q^8).\end{aligned} \tag{5.3}$$

The finite product has the same terms through q^R whenever $n > R$.

Corollary 5.2 (Reciprocal product and unrestricted partitions)**.** *For $a \neq 1$,*

$$\frac{1}{(a;q)_\infty} = \frac{1}{1-a} \sum_{r \geq 0} \left(\sum_{\lambda \vdash r} a^{\ell(\lambda)} \right) q^r. \tag{5.4}$$

In particular,

$$\frac{1}{(a;q)_\infty} = \frac{1}{1-a} \big(1 + aq + a(a+1)q^2 + a(a^2+a+1)q^3 + a(a^3+a^2+2a+1)q^4 + \mathcal{O}(q^5)\big).$$

Proof. Expand each $(1-aq^j)^{-1}$ geometrically. The multiplicity chosen from the jth factor is the multiplicity of the part j; the total power of a is the number of parts. □

Two familiar resonant specializations deserve separate mention:

$$(q;q)_\infty = \sum_{m \in \mathbb{Z}} (-1)^m q^{m(3m-1)/2} = 1 - q - q^2 + q^5 + q^7 - q^{12} - q^{15} + \cdots, \tag{5.5}$$

$$\frac{1}{(q;q)_\infty} = \sum_{r \geq 0} p(r) q^r. \tag{5.6}$$

They do not follow by naively putting $a = 1$ in (5.2), because in $(q;q)_\infty$ the first parameter moves with q.

5.2 Gaussian polynomials and rectangular partitions

Theorem 5.3 (Complete $q = 0$ expansion of a Gaussian coefficient)**.** *Let $D = k(n-k)$. Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{r=0}^{D} p_{k,n-k}(r) q^r, \tag{5.7}$$

where $p_{u,v}(r)$ is the number of partitions of r fitting in a $u \times v$ rectangle. Consequently,

$$[q^r] \begin{bmatrix} n \\ k \end{bmatrix}_q = p(r) \qquad \textit{whenever} \qquad r \leq \min(k, n-k).$$

The coefficients are palindromic: $p_{k,n-k}(r) = p_{k,n-k}(D-r)$.

Proof. The standard Ferrers-diagram interpretation gives (5.7). Every partition of r has at most r parts and largest part at most r, proving the stable range. Complementation inside the rectangle proves palindromicity. □

Thus, if the rectangle is sufficiently large,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = 1 + q + 2q^2 + 3q^3 + 5q^4 + 7q^5 + 11q^6 + \cdots$$

for as many initial terms as fit. This observation is often more useful than a quotient expansion: it gives coefficients exactly and certifies their nonnegativity.

For a q-multinomial, the coefficient of q^r counts multiset words of the prescribed content and inversion number r. Hence its constant term is one, its degree is $D(\boldsymbol{n}) = \sum_{i<j} n_i n_j$, and reciprocity makes its coefficient sequence palindromic.

5.3 The small-q expansion of Γ_q

For nonintegral x, q^x prevents an ordinary Taylor series at zero. The right replacement is a convergent generalized-power expansion.

Proposition 5.4 (Exact logarithmic expansion at $q = 0$)**.** *For $0 < q < 1$ and $x > 0$,*

$$\log \Gamma_q(x) = \sum_{m \geq 1} \frac{q^{mx}}{m(1-q^m)} - \sum_{m \geq 1} \frac{q^m}{m(1-q^m)} - (1-x) \sum_{m \geq 1} \frac{q^m}{m}. \tag{5.8}$$

All three series converge absolutely. Expanding $(1-q^m)^{-1} = \sum_{r \geq 0} q^{mr}$ gives a convergent generalized series whose logarithmic support lies in

$$\{m(x+r) : m \geq 1, r \geq 0\} \cup \mathbb{N}.$$

For rational x this is a Puiseux series; for irrational x it is naturally a Hahn-type generalized power series. In particular, the contributions from the two primitive generators of the support begin as

$$\log \Gamma_q(x) = q^x - (2-x)q + \cdots .$$

The omitted exponents lie in the displayed additive support and need not occur in increasing order relative to both x and 1; collisions cause additional cancellations at special integral values such as $x = 1, 2$.

Proof. Insert the product definition $\Gamma_q(x) = (1-q)^{1-x}(q;q)_\infty/(q^x;q)_\infty$ and use $\log(a;q)_\infty = -\sum_{m \geq 1} a^m/(m(1-q^m))$. Absolute convergence follows from $0 < q < 1$ and $x > 0$. $\square$

Chapter 6

The reciprocal point $q = \infty$ and the two real rays

6.1 Finite shifted factorials

Theorem 6.1 (Exact base reversal)**.** *For $a \neq 0$ and $n \geq 0$,*

$$(a; q)_n = (-a)^n q^{\binom{n}{2}} (a^{-1}; q^{-1})_n. \tag{6.1}$$

Consequently, with $p = q^{-1}$,

$$\begin{aligned}(a; q)_n = (1-a)(-a)^{n-1} q^{\binom{n}{2}} \Big(1 - a^{-1}p - a^{-1}p^2 \\ + (a^{-2} - a^{-1})p^3 + \mathcal{O}(p^4)\Big),\end{aligned} \tag{6.2}$$

provided $n > 3$; for smaller n the series truncates in the evident way. Every Laurent coefficient at infinity is obtained from the corresponding $q = 0$ coefficient by the substitution $a \mapsto a^{-1}$.

Proof. Factor $-aq^j$ from each factor $1 - aq^j$ and multiply. The second statement is (5.3) applied to the reversed product. □

For $q = R \to +\infty$, all powers R^{-j} retain their signs. For $q = -R \to -\infty$, the coefficient of R^{-j} acquires an additional factor $(-1)^j$, and the leading phase is $(-1)^{\binom{n}{2}}$. Thus $+\infty$ and $-\infty$ are not distinct complex singularities, but they are distinct real asymptotic rays.

The ordinary infinite product does not converge for $|q| > 1$. There is, however, a canonical renormalized finite-product limit.

Corollary 6.2 (Renormalized product outside the unit disk)**.** *If $|q| > 1$ and $a \neq 0$, then*

$$\lim_{n\to\infty} \frac{(a; q)_n}{(-a)^n q^{\binom{n}{2}}} = (a^{-1}; q^{-1})_\infty. \tag{6.3}$$

Proof. Use (6.1); the product on the right converges because $|q^{-1}| < 1$. □

6.2 Gaussian and multinomial reciprocity

Theorem 6.3 (Complete Laurent expansion of a Gaussian coefficient)**.** *With $D = k(n-k)$,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^D \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^D \sum_{r=0}^{D} p_{k,n-k}(r) q^{-r}. \tag{6.4}$$

Thus the expansion at infinity begins

$$q^D \big(1 + q^{-1} + 2q^{-2} + 3q^{-3} + \cdots \big)$$

for as long as $r \le \min(k, n-k)$. Along the negative real ray,

$$\begin{bmatrix} n \\ k \end{bmatrix}_{-R} = (-1)^D R^D \sum_{r=0}^{D} (-1)^r p_{k,n-k}(r) R^{-r}. \tag{6.5}$$

Proof. The first identity follows either from the factorial quotient or from palindromicity of the Gaussian polynomial. Insert (5.7); then substitute $q = -R$. □

The identical argument gives, for a composition $\boldsymbol{n}$,

$$\begin{bmatrix} n \\ n_1, \dots, n_s \end{bmatrix}_q = q^{D(\boldsymbol{n})} \begin{bmatrix} n \\ n_1, \dots, n_s \end{bmatrix}_q \Bigg|_{q \mapsto q^{-1}}.$$

The Laurent coefficients are the inversion numbers of multiset words read in reverse order.

For q-numbers and factorials one has the exact relations

$$[x]_q = q^{x-1} [x]_{q^{-1}} \quad \text{for integral } x, \qquad [n]_q! = q^{\binom{n}{2}} [n]_{q^{-1}}!.$$

These are the simplest instances of the same reciprocal principle.

Chapter 7

The point $q = -1$

7.1 Finite shifted factorials

Pairing even and odd exponents gives the exact value

$$(a;q)_n\Big|_{q=-1} = (1-a)^{\lceil n/2 \rceil}(1+a)^{\lfloor n/2 \rfloor}. \tag{7.1}$$

When $a \neq \pm 1$, the first derivative follows from (2.1). Writing $n = 2m$ or $2m+1$ gives

$$\frac{\mathrm{d}}{\mathrm{d}q}\log(a;q)_{2m}\Big|_{q=-1} = \frac{am((2m-1)a-1)}{1-a^2}, \tag{7.2}$$

$$\frac{\mathrm{d}}{\mathrm{d}q}\log(a;q)_{2m+1}\Big|_{q=-1} = \frac{am(1+(2m+1)a)}{1-a^2}. \tag{7.3}$$

Higher derivatives are supplied by Theorem 2.1. If $a = -1$, the odd-index factors vanish and

$$(-1;q)_n = 2^{\lceil n/2 \rceil}(2\lfloor n/2 \rfloor - 1)!!(q+1)^{\lfloor n/2 \rfloor}\big(1 + \mathcal{O}(q+1)\big), \tag{7.4}$$

where $(-1)!! = 1$. The specialization $a = 1$ is identically zero for $n \geq 1$.

7.2 q-factorials

The q-integer satisfies

$$[2r]_{-1} = 0, \qquad [2r+1]_{-1} = 1,$$

and its first nonzero term is

$$[2r]_q = r(q+1) + \mathcal{O}((q+1)^2).$$

It follows that

$$[n]_q! = \lfloor n/2 \rfloor!\,(q+1)^{\lfloor n/2 \rfloor}\big(1 + \mathcal{O}(q+1)\big). \tag{7.5}$$

This formula is a convenient way to evaluate factorial quotients at -1 without ever forming a numerical $0/0$.

7.3 The complete first jet of a Gaussian coefficient

Theorem 7.1 (Value and derivative at $q = -1$)**.** *Write*

$$n = 2a + \varepsilon, \qquad k = 2b + \delta, \qquad \varepsilon, \delta \in \{0, 1\}.$$

If $(\varepsilon, \delta) \neq (0, 1)$*, then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Bigg|_{q=-1} = \binom{a}{b}, \tag{7.6}$$

$$\frac{\mathrm{d}}{\mathrm{d}q} \begin{bmatrix} n \\ k \end{bmatrix}_q \Bigg|_{q=-1} = -\frac{k(n-k)}{2} \binom{a}{b}. \tag{7.7}$$

If $n = 2a$ *and* $k = 2b + 1$*, then*

$$\begin{bmatrix} 2a \\ 2b+1 \end{bmatrix}_q \Bigg|_{q=-1} = 0, \tag{7.8}$$

$$\frac{\mathrm{d}}{\mathrm{d}q} \begin{bmatrix} 2a \\ 2b+1 \end{bmatrix}_q \Bigg|_{q=-1} = (a-b)\binom{a}{b} = a\binom{a-1}{b}. \tag{7.9}$$

Thus every zero of a Gaussian coefficient at -1 *is simple.*

Proof. The values are the $m = 2$ case of the q-Lucas theorem; they also follow directly from (7.5) after cancelling equal powers of $q + 1$.

Assume first that the value is nonzero. The degree is $D = k(n - k)$, which is even in precisely these parity cases. Reciprocity gives $G(q) = q^D G(q^{-1})$ for $G(q) = \left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q$. Differentiating and setting $q = -1$ yields

$$G'(-1) = -DG(-1) - G'(-1),$$

so $G'(-1) = -DG(-1)/2$, proving (7.7).

Now let $n = 2a$ and $k = 2b + 1$. In

$$G(q) = \frac{(q;q)_{2a}}{(q;q)_{2b+1}(q;q)_{2a-2b-1}},$$

the numerator has a even factors, while the denominator has $b + (a - b - 1) = a - 1$ even factors. Hence one factor $q + 1$ remains. Moreover,

$$\frac{(q;q)_{2r}}{(q+1)^r} \longrightarrow 4^r r!, \qquad \frac{(q;q)_{2r+1}}{(q+1)^r} \longrightarrow 2^{2r+1} r!.$$

Taking the quotient gives

$$\lim_{q \to -1} \frac{G(q)}{q+1} = \frac{a!}{b!(a-b-1)!} = (a-b)\binom{a}{b},$$

which is the derivative at the simple zero. □

Remark 7.2 (Why the derivative formula is structurally natural). In the nonvanishing cases, palindromicity alone fixes the derivative at -1. In the vanishing case, parity forces odd degree and reciprocity only predicts that the value must vanish; the uncancelled even factorial factor supplies the missing slope. The two mechanisms together determine the entire first jet.

Proposition 7.3 (Denominator-free values over a commutative ring)**.** *Let R be a commutative ring, let $a, b \in \mathbb{N}$, and interpret ordinary binomial coefficients in R (with their usual zero extension). Then*

$$\begin{bmatrix} 2a \\ 2b \end{bmatrix}_{-1} = \binom{a}{b}, \qquad \begin{bmatrix} 2a \\ 2b+1 \end{bmatrix}_{-1} = 0,$$
$$\begin{bmatrix} 2a+1 \\ 2b \end{bmatrix}_{-1} = \binom{a}{b}, \qquad \begin{bmatrix} 2a+1 \\ 2b+1 \end{bmatrix}_{-1} = \binom{a}{b}.$$

Moreover, for every $z \in R$,

$$(z; -1)_{2a} = (1 - z^2)^a,$$
$$(z; -1)_{2a+1} = (1 - z^2)^a(1 - z).$$

These identities remain valid in positive characteristic and in rings with zero divisors.

Proof. Write $V_{n,k} = \begin{bmatrix} n \\ k \end{bmatrix}_{-1}$, with $V_{n,k} = 0$ when $k > n$. We prove the four displayed parity formulas simultaneously by induction on a. For $a = 0$, they reduce to the boundary values $V_{0,0} = V_{1,0} = V_{1,1} = 1$ and the zero extension.

For the induction step use the denominator-free q-Pascal recurrence

$$V_{n+1,k+1} = (-1)^{k+1} V_{n,k+1} + V_{n,k}.$$

Passing from row $2a + 1$ to row $2a + 2$, an even column adds two adjacent ordinary binomial coefficients and therefore gives Pascal's $\binom{a+1}{b}$; an odd column subtracts two equal coefficients and therefore vanishes. Passing once more, from row $2a + 2$ to row $2a + 3$, an even column adds the intervening zero while an odd column subtracts that zero, so both surviving entries are the same $\binom{a+1}{b}$. Together with the boundary column $V_{n,0} = 1$, these are exactly the four assertions at $a + 1$.

For the products, pair the factors with exponents $2j$ and $2j + 1$:

$$(1 - z(-1)^{2j})(1 - z(-1)^{2j+1}) = (1 - z)(1 + z) = 1 - z^2.$$

There are a such pairs in $(z; -1)_{2a}$. The product of length $2a + 1$ has in addition the even-exponent factor $1 - z$, proving both formulas. ☐

Remark 7.4 (Exact Lean crosswalk and remaining boundary)*.* The four Gaussian identities in Proposition 7.3 are the compiled declarations `Fabius.gaussianBinomial_neg_one_even_even`, `Fabius.gaussianBinomial_neg_one_even_odd_eq_zero`, `Fabius.gaussianBinomial_neg_one_odd_even`, and `Fabius.gaussianBinomial_neg_one_odd_odd`. The two product identities are `Fabius.finiteQPochhammerIn_neg_one_even` and `Fabius.finiteQPochhammerIn_neg_one_odd`. The odd-column zero lives in `FabiusFunction.QBinomialReciprocity`, where it follows from reciprocal symmetry; the other five live in `FabiusFunction.GaussianBinomialAtNegOne`. The latter module reuses the zero theorem, then needs only q-Pascal, parity, ordinary Pascal, and factor pairing to obtain the remaining values and products.

The derivative identities (7.7) and (7.9), and hence the simple-root sentence in Theorem 7.1, are proved in this report but do not yet have Lean counterparts. Their eventual formalization requires a polynomial Gaussian-coefficient interface; simplicity must be stated over the integers or in characteristic zero, because a nonzero integral slope may vanish after reduction modulo a prime.

Chapter 8

General roots of unity

8.1 Cyclotomic order and the carry criterion

Let ζ be a primitive mth root of unity. From (2.4),

$$\operatorname{ord}_{q=\zeta} \begin{bmatrix} n \\ k \end{bmatrix}_q = e_m(n,k) \in \{0,1\}. \tag{8.1}$$

Write $n = am + r$ and $k = bm + s$ with $0 \le r, s < m$. A one-line floor calculation shows

$$e_m(n,k) = 1 \quad \Longleftrightarrow \quad s > r. \tag{8.2}$$

Thus the zero records exactly the carry that occurs when the residues of k and $n-k$ are added in base m.

The q-Lucas value is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \bigg|_{q=\zeta} = \binom{a}{b} \begin{bmatrix} r \\ s \end{bmatrix}_\zeta, \tag{8.3}$$

where the right side is zero when $s > r$.

8.2 All local Taylor coefficients

Theorem 8.1 (Cyclotomic jet algorithm). *Let*

$$G(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q, \qquad e = e_m(n,k), \qquad H(q) = \frac{G(q)}{(q-\zeta)^e}.$$

Then H is holomorphic and nonzero at ζ. More explicitly,

$$H(q) = \left(\frac{\Phi_m(q)}{q-\zeta}\right)^e \prod_{\substack{d \ne m \\ e_d(n,k)=1}} \Phi_d(q). \tag{8.4}$$

If $\lambda_r = (\,\mathrm{d}^r/\,\mathrm{d}q^r) \log H(q)|_{q=\zeta}$, then

$$G(q) = (q-\zeta)^e H(\zeta) \sum_{r \ge 0} \mathcal{Y}_r(\lambda_1, \ldots, \lambda_r) \frac{(q-\zeta)^r}{r!}. \tag{8.5}$$

In particular, if $e = 1$,

$$G'(\zeta) = \Phi_m'(\zeta) \prod_{\substack{d \ne m \\ e_d(n,k)=1}} \Phi_d(\zeta). \tag{8.6}$$

Proof. Only Φ_m vanishes at a primitive mth root, and it has a simple zero there. Formula (8.4) is therefore holomorphic and nonzero. Apply Theorem 2.1 to H and restore the extracted factor. $\square$

This is an exact finite algorithm: determine the selected cyclotomic factors from floor functions, differentiate their logarithms symbolically or in the number field $\mathbb{Q}(\zeta)$, and evaluate Bell polynomials. It avoids unstable cancellation in a quotient of factorials and works to arbitrary order.

Chapter 9

Infinite shifted factorials as $q \to 1$

9.1 Fixed a: the polylogarithmic expansion

For $|q| < 1$ and $|a| < 1$, the Lambert representation is

$$\log(a; q)_\infty = -\sum_{m\geq 1} \frac{a^m}{m(1-q^m)}. \tag{9.1}$$

It is the natural starting point because the singularity as $q \to 1$ is isolated in the elementary kernel $(1-q^m)^{-1}$.

Theorem 9.1 (Fixed-parameter all-order expansion)**.** *Let $q = \mathrm{e}^{-t}$ with $t \to 0$ in a closed subsector of $\Re t > 0$. Uniformly for $|a| \leq \rho < 1$,*

$$\log(a; \mathrm{e}^{-t})_\infty \sim -\frac{\operatorname{Li}_2(a)}{t} + \frac{1}{2}\log(1-a) - \sum_{r\geq 1} \frac{B_{2r}}{(2r)!} \operatorname{Li}_{2-2r}(a) t^{2r-1}. \tag{9.2}$$

Equivalently,

$$(a; \mathrm{e}^{-t})_\infty \sim (1-a)^{1/2} \exp\left(-\frac{\operatorname{Li}_2(a)}{t}\right) \exp\left(-\sum_{r\geq 1} \frac{B_{2r}}{(2r)!} \operatorname{Li}_{2-2r}(a) t^{2r-1}\right). \tag{9.3}$$

After the dominant exponential and square-root factor are removed, every ordinary coefficient follows from complete Bell polynomials.

Proof. Use

$$\frac{1}{1-\mathrm{e}^{-x}} = \frac{1}{x} + \frac{1}{2} + \sum_{r=1}^{N-1} \frac{B_{2r}}{(2r)!} x^{2r-1} + R_N(x)$$

in (9.1). The first three kinds of terms sum to $-\operatorname{Li}_2(a)/t$, $\frac{1}{2}\log(1-a)$, and the displayed polylogarithms. For the remainder, split the m-sum at $m \leq |t|^{-1}$. On the first part, Taylor's theorem and $|a| \leq \rho$ give $\sum \rho^m m^{2N-2}|t|^{2N-1} = \mathcal{O}(|t|^{2N-1})$. The tail is exponentially small in $1/|t|$ because of ρ^m. This proves the uniform Poincare expansion. □

The first correction is

$$-\frac{t}{12} \operatorname{Li}_0(a) = -\frac{t}{12} \frac{a}{1-a},$$

and the next nonzero logarithmic correction is

$$+\frac{t^3}{720}\,\mathrm{Li}_{-2}(a) = \frac{t^3}{720}\frac{a(1+a)}{(1-a)^3}.$$

The absence of even positive powers is a property of the fixed-a regime; it is destroyed when a itself approaches one.

Formula (9.2) is classical in spirit and is closely related to the complete expansions developed by McIntosh [2]. A recent gamma-product identity of Arabi Ardehali and Rosengren gives an exact completion in which the asymptotic remainder is an absolutely convergent product of dressed gamma factors [6].

9.2 The coalescing parameter $a = q^x$

The substitution $a = \mathrm{e}^{-xt}$ lies outside every compact set $|a| \le \rho < 1$. The singularity of $\mathrm{Li}_2(a)$ at $a = 1$ interacts with $t \to 0$, producing powers of t and $\log t$ governed by the gamma function.

Theorem 9.2 (Mellin expansion of $(q^x;q)_\infty$)**.** *Let $\Re x > 0$ and $q = \mathrm{e}^{-t}$, where $t \to 0$ in a closed subsector of $|\arg t| < \pi/2$. Then*

$$\begin{aligned}\log(q^x;q)_\infty \sim\; & -\frac{\pi^2}{6t} + \left(\frac{1}{2} - x\right)\log t + \frac{1}{2}\log(2\pi) - \log\Gamma(x) \\ & + \frac{B_2(x)}{4}t - \sum_{r\ge1}\frac{B_{2r}B_{2r+1}(x)}{2r(2r+1)!}t^{2r}.\end{aligned} \tag{9.4}$$

The expansion is locally uniform for x in compact subsets of $\Re x > 0$.

Proof. For $\Re s > 1$, Mellin inversion of the exponential in the Lambert series gives

$$\log(\mathrm{e}^{-xt};\mathrm{e}^{-t})_\infty = -\frac{1}{2\pi\mathrm{i}}\int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty}\Gamma(s)\zeta(s+1)\zeta(s,x)t^{-s}\,\mathrm{d}s, \qquad c > 1. \tag{9.5}$$

The pole at $s = 1$ contributes $-\pi^2/(6t)$. At $s = 0$ there is a double pole. Using

$$\zeta(0,x) = \frac{1}{2} - x, \qquad \zeta'(0,x) = \log\Gamma(x) - \frac{1}{2}\log(2\pi)$$

gives the logarithmic and constant terms in (9.4). At $s = -1$, the residue of Γ and the values $\zeta(0) = -1/2$, $\zeta(-1,x) = -B_2(x)/2$ yield $B_2(x)t/4$. At $s = -2r$, $r \ge 1$, use

$$\mathrm{Res}_{s=-2r}\,\Gamma(s) = \frac{1}{(2r)!}, \quad \zeta(1-2r) = -\frac{B_{2r}}{2r}, \quad \zeta(-2r,x) = -\frac{B_{2r+1}(x)}{2r+1}.$$

The outer minus sign produces the coefficient in (9.4). At negative odd integers below -1, a trivial zero of $\zeta(s+1)$ cancels the gamma pole. Shifting the contour gives the stated sectorial remainder; the standard vertical estimates are uniform for x on compact subsets of the right half-plane. □

Example 9.3 (The eta specialization)**.** At $x = 1$, $B_{2r+1}(1) = 0$ for $r \ge 1$, so the algebraic expansion collapses to

$$\log(q;q)_\infty \sim -\frac{\pi^2}{6t} + \frac{1}{2}\log\frac{2\pi}{t} + \frac{t}{24}. \tag{9.6}$$

There are no further algebraic powers. The missing information is exponentially small and is recovered exactly in Chapter 11.

Example 9.4 (An integral shift)**.** For $x = 2$, $(q^2;q)_\infty = (q;q)_\infty/(1-q)$. Formula (9.4) gives

$$\log(q^2;q)_\infty \sim -\frac{\pi^2}{6t} - \frac{3}{2}\log t + \frac{1}{2}\log(2\pi) + \frac{13}{24}t - \frac{1}{24}t^2 + \cdots ,$$

which agrees term by term with subtracting $\log(1-\mathrm{e}^{-t})$ from (9.6).

Chapter 10

The q-gamma, q-beta, and q-polygamma functions

10.1 All-order expansion of Γ_q

For $0 < q < 1$,

$$\Gamma_q(x) = (1-q)^{1-x} \frac{(q;q)_\infty}{(q^x;q)_\infty}.$$

Combining Theorem 9.2 with

$$\log \frac{1-\mathrm{e}^{-t}}{t} = -\frac{t}{2} + \sum_{r\geq 1} \frac{B_{2r}}{2r(2r)!} t^{2r}$$

gives the following explicit expansion.

Theorem 10.1 (All-order $q \to 1$ expansion of Γ_q). *For $\Re x > 0$ and $q = \mathrm{e}^{-t}$,*

$$\begin{aligned} \log \Gamma_{\mathrm{e}^{-t}}(x) \sim\ & \log \Gamma(x) + \frac{(x-1)(2-x)}{4} t \\ & + \sum_{r\geq 1} \frac{B_{2r}}{2r(2r)!} \left(1 - x + \frac{B_{2r+1}(x)}{2r+1}\right) t^{2r}. \end{aligned} \tag{10.1}$$

The expansion is locally uniform on compact subsets of $\Re x > 0$ and in closed subsectors of $|\arg t| < \pi/2$.

Proof. Subtract (9.4) from its $x = 1$ specialization and add $(1-x)\log(1-\mathrm{e}^{-t})$. The t^{-1}, $\log(2\pi)$, and logarithmic terms cancel. The linear coefficient simplifies to

$$\frac{1}{24} - \frac{B_2(x)}{4} - \frac{1-x}{2} = \frac{(x-1)(2-x)}{4}.$$

For $r \geq 1$, the $x = 1$ Bernoulli-polynomial term vanishes, and the two remaining contributions combine exactly as displayed. □

The first terms are

$$\begin{aligned} \log \Gamma_q(x) =\ & \log \Gamma(x) + \frac{(x-1)(2-x)}{4} t \\ & + \frac{1}{24}\left(1 - x + \frac{B_3(x)}{3}\right) t^2 - \frac{1}{2880}\left(1 - x + \frac{B_5(x)}{5}\right) t^4 + \mathcal{O}(t^6). \end{aligned} \tag{10.2}$$

For $x = 3$, this reduces to

$$\log \Gamma_q(3) = \log(1 + \mathrm{e}^{-t}) = \log 2 - \frac{t}{2} + \frac{t^2}{8} - \frac{t^4}{192} + \cdots ,$$

providing an exact consistency check.

10.2 q-digamma and higher logarithmic derivatives

Termwise differentiation in x is legitimate on compact subsets of the right half-plane.

Corollary 10.2 (Expansion of the q-digamma function). *Let $\psi_q(x) = \partial_x \log \Gamma_q(x)$. Then*

$$\begin{aligned}\psi_{\mathrm{e}^{-t}}(x) \sim \psi(x) + \frac{3-2x}{4}t \\ + \sum_{r\geq 1} \frac{B_{2r}}{2r(2r)!}\big(B_{2r}(x) - 1\big)t^{2r}.\end{aligned} \tag{10.3}$$

Further x-derivatives follow from $B_n'(x) = nB_{n-1}(x)$.

10.3 The q-beta function

Since $B_q(x,y) = \Gamma_q(x)\Gamma_q(y)/\Gamma_q(x+y)$, define

$$c_1(x) = \frac{(x-1)(2-x)}{4}, \qquad c_{2r}(x) = \frac{B_{2r}}{2r(2r)!}\left(1 - x + \frac{B_{2r+1}(x)}{2r+1}\right).$$

Then

$$\log \frac{B_{\mathrm{e}^{-t}}(x,y)}{B(x,y)} \sim \big(c_1(x) + c_1(y) - c_1(x+y)\big)t + \sum_{r\geq 1}\big(c_{2r}(x) + c_{2r}(y) - c_{2r}(x+y)\big)t^{2r}. \tag{10.4}$$

The linear coefficient simplifies to

$$c_1(x) + c_1(y) - c_1(x+y) = \frac{xy-1}{2}. \tag{10.5}$$

Bell polynomials exponentiate (10.4) to any desired order.

Chapter 11

Modular and exponentially improved expansions

11.1 The eta product at $q = 1$

Dedekind eta modularity gives an exact completion of (9.6):

Theorem 11.1 (Exact modular completion)**.** *For $t > 0$,*

$$(\mathrm{e}^{-t}; \mathrm{e}^{-t})_\infty = \sqrt{\frac{2\pi}{t}} \exp\left(-\frac{\pi^2}{6t} + \frac{t}{24}\right) (\mathrm{e}^{-4\pi^2/t}; \mathrm{e}^{-4\pi^2/t})_\infty. \tag{11.1}$$

In particular, truncating after the displayed elementary factor incurs a relative error $\mathcal{O}(\mathrm{e}^{-4\pi^2/t})$.

Proof. Apply $\eta(-1/\tau) = \sqrt{-\mathrm{i}\tau}\,\eta(\tau)$ with $\tau = \mathrm{i}t/(2\pi)$ and use $\eta(\tau) = q^{1/24}(q;q)_\infty$. ☐

This exact identity explains why every algebraic coefficient beyond $t/24$ vanishes: the remainder is not zero, but beyond all orders.

11.2 The eta product on the negative real ray

Let $x = \mathrm{e}^{-t}$. Separating even and odd indices gives

$$(-x; -x)_\infty = \frac{(x^2; x^2)_\infty^3}{(x; x)_\infty (x^4; x^4)_\infty}. \tag{11.2}$$

Applying (11.1) at $t, 2t, 4t$ yields another exact exponentially improved formula.

Theorem 11.2 (Exact completion at $q = -1$)**.** *For $t > 0$, set $Q_c = \exp(-c\pi^2/t)$. Then*

$$(-\mathrm{e}^{-t}; -\mathrm{e}^{-t})_\infty = \sqrt{\frac{\pi}{t}} \exp\left(-\frac{\pi^2}{24t} + \frac{t}{24}\right) \frac{(Q_2; Q_2)_\infty^3}{(Q_4; Q_4)_\infty (Q_1; Q_1)_\infty}. \tag{11.3}$$

Consequently,

$$(-\mathrm{e}^{-t}; -\mathrm{e}^{-t})_\infty = \sqrt{\frac{\pi}{t}} \exp\left(-\frac{\pi^2}{24t} + \frac{t}{24}\right) \left(1 + \mathcal{O}(\mathrm{e}^{-\pi^2/t})\right). \tag{11.4}$$

Proof. Identity (11.2) follows from

$$(-x; x^2)_\infty = \frac{(x^2; x^4)_\infty}{(x; x^2)_\infty}, \quad (x; x^2)_\infty = \frac{(x; x)_\infty}{(x^2; x^2)_\infty}, \quad (x^2; x^4)_\infty = \frac{(x^2; x^2)_\infty}{(x^4; x^4)_\infty}.$$

Substitute the three modular completions and simplify the elementary factors. ☐

11.3 An arbitrary eta cusp

Let h, m be coprime, $m > 0$, and

$$\tau = \frac{h}{m} + \frac{\mathrm{i}t}{2\pi}, \qquad q = \mathrm{e}^{2\pi\mathrm{i}\tau} = \mathrm{e}^{2\pi\mathrm{i}h/m}\mathrm{e}^{-t}.$$

Choose integers a, b such that $-ah - bm = 1$, and let $\gamma = \left(\begin{smallmatrix} a & b \\ m & -h \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Dedekind eta transformation gives the exact identity

$$(q;q)_\infty = \mathrm{e}^{-\pi\mathrm{i}\tau/12}\,\epsilon(\gamma)^{-1}(m\tau - h)^{-1/2}\mathrm{e}^{\pi\mathrm{i}\gamma\tau/12}\big(\mathrm{e}^{2\pi\mathrm{i}\gamma\tau};\mathrm{e}^{2\pi\mathrm{i}\gamma\tau}\big)_\infty, \tag{11.5}$$

where $\epsilon(\gamma)$ is the eta multiplier and a consistent square-root branch is understood. Since

$$\gamma\tau = \frac{a}{m} + \frac{2\pi\mathrm{i}}{m^2 t}.$$

After the chosen Bezout relation is used, the final product differs from one by $\mathcal{O}(\mathrm{e}^{-4\pi^2/(m^2t)})$. In particular,

$$|(q;q)_\infty| \sim \sqrt{\frac{2\pi}{mt}}\exp\left(-\frac{\pi^2}{6m^2t} + \frac{t}{24}\right). \tag{11.6}$$

The cases $m = 1$ and $m = 2$ reproduce (11.1) and (11.4).

11.4 A theta transseries by Poisson summation

For

$$\Theta(z;q) = \sum_{n\in\mathbb{Z}} q^{n(n-1)/2}z^n, \qquad z = \mathrm{e}^u,$$

completion of the square and Poisson summation give, for $t > 0$,

$$\Theta(\mathrm{e}^u;\mathrm{e}^{-t}) = \sqrt{\frac{2\pi}{t}}\exp\left(\frac{(u+t/2)^2}{2t}\right)\sum_{k\in\mathbb{Z}}\exp\left(-\frac{2\pi^2k^2}{t} + \frac{2\pi\mathrm{i}k(u+t/2)}{t}\right). \tag{11.7}$$

This is an exact transseries rather than a merely formal power expansion. Its dominant exponential changes when u crosses Stokes lines, and multiplication by the Jacobi triple product converts the statement into coupled asymptotics of three infinite q-Pochhammer factors.

Chapter 12

Radial asymptotics at arbitrary roots of unity

12.1 A fixed nonresonant parameter

Let ζ be a primitive mth root of unity and approach it radially by

$$q = \zeta \mathrm{e}^{-t}, \qquad t \to 0, \qquad \Re t > 0.$$

For fixed $|a| < 1$, the Lambert series can be separated into indices divisible by m and indices not divisible by m. The former create the singular dilogarithmic scale; the latter are regular but remember the phase ζ.

Definition 12.1 (Cyclic quantum dilogarithm). Set

$$D_\zeta(a) = \prod_{s=1}^{m-1} (1 - \zeta^s a)^s. \tag{12.1}$$

For $m = 1$ the empty product is one.

Define rational functions

$$\mathcal{E}_j(u) = \left(u \frac{\mathrm{d}}{\mathrm{d}u} \right)^j \frac{1}{1-u}. \tag{12.2}$$

Thus

$$\mathcal{E}_0(u) = \frac{1}{1-u}, \quad \mathcal{E}_1(u) = \frac{u}{(1-u)^2}, \quad \mathcal{E}_2(u) = \frac{u(1+u)}{(1-u)^3},$$

and $\mathcal{E}_j(u) = \mathrm{Li}_{-j}(u)$ for $j \geq 1$, interpreted as a rational function when $u \neq 1$.

Theorem 12.2 (All-order fixed-a root-of-unity expansion). *Let $|a| \leq \rho < 1$ and let ζ be primitive of order m. Then, uniformly for a in that disk and for t in a closed subsector of $\Re t > 0$,*

$$\begin{aligned} \log(a; \zeta \mathrm{e}^{-t})_\infty \sim & -\frac{\mathrm{Li}_2(a^m)}{m^2 t} + \frac{1}{2} \log(1 - a^m) - \frac{1}{m} \log D_\zeta(a) \\ & + \sum_{j \geq 1} C_j(a, \zeta) t^j, \end{aligned} \tag{12.3}$$

where

$$C_j(a, \zeta) = \frac{(-1)^{j+1}}{j!} \sum_{\substack{r \geq 1 \\ m \nmid r}} a^r r^{j-1} \mathcal{E}_j(\zeta^r)$$

$$- \frac{B_{j+1} m^{j-1}}{(j+1)!} \operatorname{Li}_{1-j}(a^m). \tag{12.4}$$

All coefficient sums converge absolutely. For $m = 1$, the first line involving D_ζ is empty and (12.3) reduces to Theorem 9.1.

Proof. Start from

$$L(t) = -\sum_{r\geq 1} \frac{a^r}{r(1 - \zeta^r e^{-rt})}.$$

When $r = m\ell$, expand

$$\frac{1}{1 - e^{-m\ell t}} = \frac{1}{m\ell t} + \frac{1}{2} + \sum_{j\geq 1} \frac{B_{j+1}}{(j+1)!} (m\ell t)^j.$$

The singular, constant, and t^j contributions are respectively

$$-\frac{\operatorname{Li}_2(a^m)}{m^2 t}, \qquad \frac{1}{2m} \log(1 - a^m), \qquad -\frac{B_{j+1} m^{j-1}}{(j+1)!} \operatorname{Li}_{1-j}(a^m) t^j.$$

For $m \nmid r$, the dilation identity

$$\frac{1}{1 - u e^{-x}} = \sum_{j\geq 0} \frac{(-x)^j}{j!} \mathcal{E}_j(u)$$

gives the first sum in (12.4). Its constant term is

$$-\sum_{\substack{r\geq 1\\ m\nmid r}} \frac{a^r}{r(1-\zeta^r)}.$$

To simplify it, observe that for $m \nmid r$,

$$\sum_{s=1}^{m-1} s\zeta^{sr} = -\frac{m}{1-\zeta^r},$$

while for $m \mid r$ the sum is $m(m-1)/2$. Expanding $\log D_\zeta(a)$ from (12.1) therefore yields

$$-\sum_{\substack{r\geq 1\\ m\nmid r}} \frac{a^r}{r(1-\zeta^r)} = -\frac{1}{m} \log D_\zeta(a) + \frac{m-1}{2m} \log(1 - a^m).$$

Adding the constant from the divisible indices gives the constant in (12.3).

For the remainder estimate, choose $\delta > 0$ smaller than the distance from 0 to every singularity of $x \mapsto (1 - ue^{-x})^{-1}$ with $u \in \{\zeta, \dots, \zeta^{m-1}\}$, and also choose $\delta < 2\pi$ for the resonant Bernoulli expansion. Split the Lambert sum at $r \leq \delta/|t|$. On this first range the relevant Taylor remainders are uniformly bounded by a polynomial in r times the requested power of t. On the complementary range, $\rho^r = \mathcal{O}(\exp(-c/|t|))$ for some $c > 0$, so the tail is smaller than every algebraic power of t. This is the same small-range/exponentially-small-tail argument as in Theorem 9.1. □

Remark 12.3 (Normalization used in the roots-of-unity literature). If one writes $q = \zeta e^{-\varepsilon/m}$ instead, then $t = \varepsilon/m$ and the leading term becomes $-\operatorname{Li}_2(a^m)/(m\varepsilon)$. This is the normalization commonly used in radial asymptotics of Nahm sums. The cyclic factor $D_\zeta(a)^{-1/m}$ is exactly the finite root-of-unity datum that survives after the singular dilogarithm is removed; compare Garoufalidis–Zagier [5].

12.2 The first coefficients

Formula (12.4) is already computationally explicit. For example,

$$C_1 = \sum_{m\nmid r} a^r \frac{\zeta^r}{(1-\zeta^r)^2} - \frac{1}{12}\,\mathrm{Li}_0(a^m),$$
$$C_2 = -\frac{1}{2}\sum_{m\nmid r} a^r r \frac{\zeta^r(1+\zeta^r)}{(1-\zeta^r)^3},$$

where the B_3 contribution vanishes. The sums may be reduced to finite linear combinations of $\mathrm{Li}_{1-j}(a\zeta^s)$ by discrete Fourier analysis. Keeping the residue-class sums is often numerically better conditioned and makes the distinction between resonant and nonresonant indices transparent.

12.3 Coalescence and why a second scaling is necessary

The hypothesis $|a| \le \rho < 1$ excludes $a \to \zeta^{-r}$, where one of the finite cyclic factors vanishes. It also excludes the eta specialization $a = q$, for which $a^m \to 1$. In those regimes the dilogarithm in (12.3) develops a logarithmic singularity that must be combined with the cyclic factor before expansion. The resulting scale contains powers of t, $\log t$, Bernoulli polynomials in the shift parameter, and an exponentially small completion. General all-order shifted formulae of this kind underlie the root-of-unity analysis of Nahm sums [5]; the eta specialization is treated exactly by (11.5).

A central open problem, formulated more sharply in Chapter 18, is to construct one uniform normal form that contains both the fixed-a expansion (12.3) and the coalescing $a^m \to 1$ expansion without separate case distinctions.

Chapter 13

Other q-analogues near $q = 1$

13.1 The two standard q-exponentials

Use the normalization in the source monograph,

$$e_q(x) = \sum_{n\geq 0} \frac{x^n}{[n]_q!} = \frac{1}{((1-q)x;q)_\infty}, \qquad E_q(x) = \sum_{n\geq 0} \frac{q^{\binom{n}{2}} x^n}{[n]_q!} = (-(1-q)x;q)_\infty.$$

Let $h = 1 - q$. Their logarithms have the exact convergent representations

$$\log e_q(x) = \sum_{m\geq 1} \frac{h^m x^m}{m(1-(1-h)^m)}, \tag{13.1}$$

$$\log E_q(x) = \sum_{m\geq 1} \frac{(-1)^{m+1} h^m x^m}{m(1-(1-h)^m)}. \tag{13.2}$$

At any prescribed order in h, only finitely many m contribute. Consequently these equations are also coefficient-generation algorithms. Expanding gives

$$\log e_q(x) = x + \frac{x^2}{4}h + \left(\frac{x^2}{8} + \frac{x^3}{9}\right)h^2 + \frac{x^2(9x^2+16x+9)}{144}h^3 + \mathcal{O}(h^4), \tag{13.3}$$

$$\log E_q(x) = x - \frac{x^2}{4}h + \left(-\frac{x^2}{8} + \frac{x^3}{9}\right)h^2 - \frac{x^2(9x^2-16x+9)}{144}h^3 + \mathcal{O}(h^4). \tag{13.4}$$

Bell exponentiation converts these logarithmic formulae to expansions of the functions themselves. The relation $e_q(x)E_q(-x) = 1$ is visible term by term.

Derivation of (13.1)–(13.2). Apply the Lambert identity (9.1) with $a = \mp hx$ and $q = 1 - h$, remembering the minus sign for the reciprocal in e_q. □

13.2 The q-difference operator as an operator-valued q-number

Let $\Theta_x = x\frac{\mathrm{d}}{\mathrm{d}x}$. For $q = \mathrm{e}^{-t}$,

$$D_q f(x) = \frac{f(x) - f(qx)}{(1-q)x} = x^{-1}\frac{1-\mathrm{e}^{-t\Theta_x}}{1-\mathrm{e}^{-t}} f(x) = x^{-1}[\Theta_x]_{\mathrm{e}^{-t}} f(x). \tag{13.5}$$

Using (3.6) with the scalar x replaced by the commuting operator Θ_x gives an all-order expansion. The first terms are

$$D_{e^{-t}} = x^{-1}\Big[\Theta_x - \frac{t}{2}\Theta_x(\Theta_x - 1) + \frac{t^2}{12}\Theta_x(\Theta_x - 1)(2\Theta_x - 1) - \frac{t^3}{24}\Theta_x^2(\Theta_x - 1)^2 + \mathcal{O}(t^4)\Big]. \tag{13.6}$$

Since $x^{-1}\Theta_x = \mathrm{d}/\mathrm{d}x$, the first term is the classical derivative. Formula (13.6) is valid on polynomials as a finite operator identity and on analytic functions under the usual local convergence conditions.

13.3 Jackson integration by Euler–Maclaurin

For $q = e^{-t}$,

$$\int_0^a f(x)\,\mathrm{d}_q x = (1 - e^{-t})a \sum_{n\geq 0} e^{-nt} f(ae^{-nt}).$$

Put $g(s) = ae^{-s} f(ae^{-s})$. Suppose that f is sufficiently smooth on $[0, a]$ and that the transformed derivatives of g required below are integrable and decay at infinity. Euler–Maclaurin then gives

$$t\sum_{n\geq 0} g(nt) \sim \int_0^\infty g(s)\,\mathrm{d}s + \frac{t}{2}g(0) - \sum_{r\geq 1}\frac{B_{2r}t^{2r}}{(2r)!}g^{(2r-1)}(0). \tag{13.7}$$

Multiplication by $(1 - e^{-t})/t$ yields

$$\int_0^a f(x)\,\mathrm{d}_{e^{-t}} x = \int_0^a f(x)\,\mathrm{d}x + \frac{t}{2}\left(af(a) - \int_0^a f(x)\,\mathrm{d}x\right) + t^2\left(\frac{a^2 f'(a)}{12} - \frac{af(a)}{6} + \frac{1}{6}\int_0^a f(x)\,\mathrm{d}x\right) + \mathcal{O}(t^3). \tag{13.8}$$

The full expansion is an explicit endpoint differential series. It provides a systematic bridge between Jackson q-integrals and ordinary quadrature, and is particularly convenient for perturbing q-beta integrals.

Chapter 14

Basic hypergeometric series as differential-operator deformations

14.1 A coefficientwise all-order operator

Consider the balanced-length family

$$F_q(z) = {}_r\phi_{r-1}\left[\begin{matrix} q^{\alpha_1}, \dots, q^{\alpha_r} \\ q^{\beta_1}, \dots, q^{\beta_{r-1}} \end{matrix}; q, z\right], \tag{14.1}$$

with $\beta_r = 1$ added to account for the factor $(q;q)_n$. Its classical limit is

$$F(z) = {}_rF_{r-1}\left[\begin{matrix} \alpha_1, \dots, \alpha_r \\ \beta_1, \dots, \beta_{r-1} \end{matrix}; z\right].$$

Let $\theta = z\frac{\mathrm{d}}{\mathrm{d}z}$ and define the polynomial

$$\begin{aligned} P_{2s}(X) = \sum_{i=1}^{r} &\frac{B_{2s+1}(\alpha_i + X) - B_{2s+1}(\alpha_i)}{2s+1} \\ &- \sum_{j=1}^{r} \frac{B_{2s+1}(\beta_j + X) - B_{2s+1}(\beta_j)}{2s+1}. \end{aligned} \tag{14.2}$$

Because the two parameter lists have equal length, the leading X^{2s+1} terms cancel, so P_{2s} has degree at most $2s$.

Theorem 14.1 (Formal differential-operator expansion). *Let*

$$A = \sum_{i=1}^{r} \alpha_i - \sum_{j=1}^{r} \beta_j.$$

Then, coefficientwise in z and asymptotically as $q = \mathrm{e}^{-t} \to 1$,

$$F_q(z) \sim \exp\left(-\frac{A}{2} t\theta + \sum_{s\geq 1} \frac{B_{2s}}{2s(2s)!} P_{2s}(\theta) t^{2s}\right) F(z). \tag{14.3}$$

On every compact disk on which the defining series and a sufficient number of its parameterwise majorants converge uniformly, the expansion may be differentiated and summed termwise to any fixed order.

Proof. The coefficient of z^n in (14.1) is a product of ratios $(q^\alpha; q)_n/(q^\beta; q)_n$. Apply Theorem 3.2. The linear difference is An, and the $2s$th power sum difference is exactly $P_{2s}(n)$. Since $\theta z^n = nz^n$, replacing n by θ packages the coefficientwise multiplier into the operator exponential (14.3). □

14.2 The first two perturbative orders

Put

$$B = \sum_{i=1}^{r} \alpha_i^2 - \sum_{j=1}^{r} \beta_j^2.$$

Since $P_2(n) = An(n-1) + Bn$, expansion of the operator exponential gives

$$\begin{aligned} F_q(z) = F(z) - \frac{A}{2} t\, \theta F(z) \\ + \frac{t^2}{24} \left((3A^2 + A)\theta^2 + (B - A)\theta \right) F(z) + \mathcal{O}(t^3). \end{aligned} \tag{14.4}$$

If $A = 0$, the entire first-order correction vanishes. If also $B = 0$, the expansion starts at order t^4 in the logarithmic operator and at order t^4 in the function itself. These cancellation criteria offer a quick way to identify especially faithful q-deformations of classical hypergeometric functions.

Example 14.2 (q-binomial theorem)**.** For ${}_1\phi_0(q^\alpha; -; q, z)$, one has $F(z) = (1 - z)^{-\alpha}$, $A = \alpha - 1$, and $B = \alpha^2 - 1$. Formula (14.4) gives the first corrections entirely in terms of $\theta(1 - z)^{-\alpha}$ and $\theta^2(1 - z)^{-\alpha}$, without re-expanding the infinite product quotient separately.

14.3 Terminating series and roots of unity

For a terminating basic hypergeometric series, each summand is a rational function of q and the parameters. At a chosen root of unity, cyclotomic factorization decides whether apparent zero-over-zero factors cancel, whether the summand has a genuine zero, or whether it has a genuine pole. The reliable procedure is therefore:

1. factor every finite shifted factorial into cyclotomic factors in the numerator and denominator;
2. compare the local valuations and extract the resulting power of $q - \zeta$ (negative powers signal an actual pole rather than a removable singularity);
3. evaluate the regular nonzero quotient in $\mathbb{Q}(\zeta)$;
4. use Bell polynomials for its higher jet and then sum the finitely many terms whenever the complete terminating expression is regular.

Direct numerical substitution into an uncancelled quotient can lose every significant digit. The cyclotomic method computes the exact local algebra first and only then performs numerical evaluation.

Chapter 15

A double-scaling expansion for Gaussian coefficients

15.1 Why fixed n is not uniform

The convergent fixed-n expansion (4.1) has radius $2\pi/n$ in t. It is therefore not uniform when $n \to \infty$. The natural transition coordinate is

$$q = \mathrm{e}^{-\tau/N}, \qquad k = \alpha N, \qquad 0 < \alpha < 1,$$

with τ fixed. This regime interpolates between the classical binomial coefficient ($\tau \to 0$) and a genuinely dilogarithmic q-deformation.

Define

$$I_\tau(\beta) = \int_0^\beta \log(1 - \mathrm{e}^{-\tau x})\,\mathrm{d}x = \frac{\mathrm{Li}_2(\mathrm{e}^{-\beta\tau}) - \pi^2/6}{\tau}. \tag{15.1}$$

15.2 A finite q-factorial with a moving endpoint

Theorem 15.1 (All-order moving-endpoint expansion)**.** *Fix $\tau > 0$ and $\beta > 0$, and let βN be integral. With $q = \mathrm{e}^{-\tau/N}$,*

$$\begin{aligned}\log(q;q)_{\beta N} \sim N I_\tau(\beta) + \frac{1}{2}\log\left(\frac{2\pi N(1-\mathrm{e}^{-\beta\tau})}{\tau}\right) + \frac{\tau}{24N}\coth\left(\frac{\beta\tau}{2}\right) \\ + \sum_{r\geq 2} \frac{B_{2r}\tau^{2r-1}}{(2r)!N^{2r-1}}\,\mathrm{Li}_{2-2r}(\mathrm{e}^{-\beta\tau}).\end{aligned} \tag{15.2}$$

The expansion is uniform when (β,τ) ranges over a compact subset of $(0,\infty)^2$. After the N^{-1} term, only odd powers of N^{-1} occur.

Proof. Write

$$\log(1-\mathrm{e}^{-\tau x}) = \log(\tau x) + g(\tau x), \qquad g(u) = \log\frac{1-\mathrm{e}^{-u}}{u}.$$

Then

$$\log(q;q)_{\beta N} = \beta N \log\frac{\tau}{N} + \log(\beta N)! + \sum_{j=1}^{\beta N} g\left(\frac{\tau j}{N}\right).$$

Apply Stirling's expansion to the factorial and Euler–Maclaurin to the smooth last sum. Their leading terms combine into $NI_\tau(\beta)$. The endpoint terms simplify to the logarithm in (15.2).

For completeness, let $H(x) = g(\tau x)$. The coefficient of $N^{-(2r-1)}$ before simplification is

$$\frac{B_{2r}}{(2r)!}\left(\frac{(2r-2)!}{\beta^{2r-1}} + H^{(2r-1)}(\beta) - H^{(2r-1)}(0)\right). \tag{15.3}$$

For $r = 1$, $H'(0) = -\tau/2$, and (15.3) becomes $\tau\coth(\beta\tau/2)/24$. For $r \geq 2$, $H^{(2r-1)}(0) = 0$ and

$$H^{(2r-1)}(\beta) = \tau^{2r-1}\operatorname{Li}_{2-2r}(\mathrm{e}^{-\beta\tau}) - \frac{(2r-2)!}{\beta^{2r-1}}.$$

The rational endpoint term cancels the Stirling term, leaving exactly the polylogarithm in (15.2). Standard uniform Euler–Maclaurin bounds prove the stated uniformity. □

15.3 Gaussian coefficients

Define the deformed entropy

$$\begin{aligned}\Phi(\alpha,\tau) &:= I_\tau(1) - I_\tau(\alpha) - I_\tau(1-\alpha)\\ &= \frac{\operatorname{Li}_2(\mathrm{e}^{-\tau}) - \operatorname{Li}_2(\mathrm{e}^{-\alpha\tau}) - \operatorname{Li}_2(\mathrm{e}^{-(1-\alpha)\tau}) + \pi^2/6}{\tau}.\end{aligned} \tag{15.4}$$

Also put

$$A_1(\beta;\tau) = \frac{\tau}{24}\coth\left(\frac{\beta\tau}{2}\right), \tag{15.5}$$

$$A_{2r-1}(\beta;\tau) = \frac{B_{2r}\tau^{2r-1}}{(2r)!}\operatorname{Li}_{2-2r}(\mathrm{e}^{-\beta\tau}) \qquad (r \geq 2). \tag{15.6}$$

Corollary 15.2 (All-order double scaling of $\left[{N \atop k}\right]_q$). *Let $k = \alpha N$ be integral, with (α,τ) in a compact subset of $(0,1)\times(0,\infty)$. Then*

$$\begin{aligned}\log\begin{bmatrix}N\\ \alpha N\end{bmatrix}_{\mathrm{e}^{-\tau/N}} &\sim N\Phi(\alpha,\tau) - \frac{1}{2}\log\frac{2\pi N}{\tau}\\ &\quad+\frac{1}{2}\log\frac{1-\mathrm{e}^{-\tau}}{(1-\mathrm{e}^{-\alpha\tau})(1-\mathrm{e}^{-(1-\alpha)\tau})}\\ &\quad+\sum_{r\geq 1}\frac{A_{2r-1}(1;\tau) - A_{2r-1}(\alpha;\tau) - A_{2r-1}(1-\alpha;\tau)}{N^{2r-1}}.\end{aligned} \tag{15.7}$$

Proof. Subtract the two denominator instances of (15.2) from the numerator instance. □

The first correction is therefore

$$\frac{\tau}{24N}\left[\coth\frac{\tau}{2} - \coth\frac{\alpha\tau}{2} - \coth\frac{(1-\alpha)\tau}{2}\right]. \tag{15.8}$$

The next is

$$-\frac{\tau^3}{720N^3}\left[\operatorname{Li}_{-2}(\mathrm{e}^{-\tau}) - \operatorname{Li}_{-2}(\mathrm{e}^{-\alpha\tau}) - \operatorname{Li}_{-2}(\mathrm{e}^{-(1-\alpha)\tau})\right]. \tag{15.9}$$

15.4 Recovery of ordinary Stirling asymptotics

As $\tau \to 0$,

$$\Phi(\alpha, \tau) \longrightarrow -\alpha \log \alpha - (1-\alpha)\log(1-\alpha).$$

The logarithmic prefactor in (15.7) tends to

$$-\frac{1}{2}\log\bigl(2\pi N\alpha(1-\alpha)\bigr),$$

and (15.8) tends to

$$\frac{1}{12N}\left(1 - \frac{1}{\alpha} - \frac{1}{1-\alpha}\right).$$

These are exactly the entropy, Gaussian prefactor, and first Stirling correction for $\binom{N}{\alpha N}$. Hence the double-scaling formula is a uniform q-deformation of classical binomial asymptotics rather than a separate limit formula.

15.5 Multinomial double scaling

Let $\alpha_1 + \cdots + \alpha_s = 1$ with every $\alpha_i > 0$ and $n_i = \alpha_i N$ integral. The same subtraction gives

$$\begin{aligned}\log \begin{bmatrix} N \\ \alpha_1 N, \ldots, \alpha_s N \end{bmatrix}_q \Bigg|_{q=\mathrm{e}^{-\tau/N}} \sim N\left(I_\tau(1) - \sum_{i=1}^{s} I_\tau(\alpha_i)\right) \\ + \frac{1-s}{2}\log\frac{2\pi N}{\tau} + \frac{1}{2}\log\frac{1-\mathrm{e}^{-\tau}}{\prod_i(1-\mathrm{e}^{-\alpha_i\tau})} \\ + \sum_{r\geq 1}\frac{A_{2r-1}(1;\tau) - \sum_i A_{2r-1}(\alpha_i;\tau)}{N^{2r-1}}.\end{aligned} \tag{15.10}$$

The formula is uniform when the composition remains in a compact subset of the open simplex.

Chapter 16

Implications for the ProveIt and Fabius-function programme

The primary purpose of this report is the q-series expansion problem itself, but several outputs are directly reusable in the surrounding Fabius-function research.

16.1 Effective coefficient generation

Every finite expansion in the report reduces to operations already friendly to formalization and exact computation:

(a) finite power sums and Bernoulli polynomials;

(b) cyclotomic factorization over $\mathbb{Z}[q]$;

(c) logarithmic derivatives of nonvanishing polynomials;

(d) complete Bell polynomials or the equivalent triangular recurrence;

(e) exact rational or cyclotomic-number arithmetic.

Thus local sensitivity of a formula involving q-binomial coefficients at the dyadic points $q = 1/2$ or $q = 2$ can be computed by Theorems 2.1 and 2.4, while a deformation of the base toward one can use the much smaller Bernoulli data in Theorem 4.1. No numerical differentiation is required.

16.2 Reciprocal dyadic normalizations

The Fabius and Rvachev formulae frequently contain powers of two together with Gaussian coefficients or products at bases 2 and $1/2$. Identities (6.1) and (6.4) make the normalization explicit before numerical evaluation. Moving a large power of q outside the polynomial leaves a convergent series in q^{-1} and prevents overflow or catastrophic cancellation. The same transformation is useful in formal proofs because it replaces a statement at $q > 1$ by one inside the convergence disk.

16.3 Moment and cumulant organization

Gaussian coefficients occurring in moment formulae can be expanded in a centered form whose odd logarithmic coefficients vanish. The exact cumulants in Theorem 4.3 provide a compact replacement for repeated ad hoc differentiation. In particular, when a Fabius-related expression contains a weighted sum of Gaussian coefficients near $q = 1$, its perturbative coefficients can often be organized as finite combinations of inversion cumulants and hence of the factored polynomials (4.5)–(4.7).

16.4 Root-of-unity cancellation as exact algebra

Alternating signs and Thue–Morse structures naturally bring $q = -1$ and other roots of unity into intermediate formulae. The cyclotomic-jet method separates genuine zeros from removable factorial quotients. The closed first jet in Theorem 7.1 is especially useful when a limiting formula contains one vanishing Gaussian factor: the limit is then determined by a single integer binomial coefficient rather than by a symbolic l'Hopital calculation.

Chapter 17

Numerical validation and reproducible computation

17.1 What was checked

The companion program `q_expansion_experiments.py` uses exact integer polynomial arithmetic for finite Gaussian coefficients and 80-digit `mpmath` arithmetic for infinite products. It performs the following checks.

1. It derives $\Delta_2, \Delta_4, \Delta_6$ symbolically and reduces symmetric expressions to polynomials in n and $D = k(n-k)$.

2. It expands $\prod_{j\geq 1}(1 - aq^j)$ through a requested degree and verifies the weighted distinct-partition coefficients at $q = 0$.

3. It constructs every $\begin{bmatrix} n \\ k \end{bmatrix}_q$ with $0 \leq k \leq n \leq 20$ by Pascal recursion, checks the normalized $q = 1$ expansion through fourth order, and checks both formulas in Theorem 7.1. Each suite covers all 230 index pairs and passes exactly.

4. It evaluates the Mellin expansion (9.4), the q-gamma expansion (10.1), and the root-of-unity expansion (12.3) against directly summed logarithmic products.

5. It compares the double-scaling formula (15.7) through N^{-1} and through N^{-3}.

17.2 Representative errors

Table 17.1: High-precision checks produced by the companion program. Errors are absolute errors in logarithms, except in the final two rows where they are signed.

Formula and parameters	Truncation	Error
$(q^x; q)_\infty$, $x = 2.5$, $t = 0.1$	through t^6	2.65×10^{-14}
$\Gamma_q(x)$, $x = 3.7$, $t = 0.08$	through t^6	5.01×10^{-13}
root $m = 3$, $a = 0.2$, $t = 0.05$	through t^6	3.48×10^{-11}
$N = 100$, $\alpha = 0.4$, $\tau = 1.7$	through N^{-1}	$+5.35 \times 10^{-8}$
same double scaling	through N^{-3}	-8.69×10^{-12}

The root-of-unity error is larger than the two positive-real errors because it combines complex logarithm branches and a truncated residue-class coefficient sum. Its observed decrease under smaller t and higher truncation is consistent with the predicted next power. At the displayed double-scaling test point, adding the explicit N^{-3} term reduces the error by roughly four orders of magnitude and leaves a remainder consistent with the next N^{-5} term.

17.3 Stable evaluation rules

The experiments suggest the following implementation discipline.

1. Evaluate logarithms of long products and exponentiate only at the end.
2. Near $q = 1$, use $t = -\log q$ computed by a compensated `log1p`; use `expm1` for $1 - q^j$.
3. Near a root of unity, cancel cyclotomic zero factors symbolically before floating-point evaluation.
4. At $|q| > 1$, reverse the base and remove the explicit power of q before evaluation.
5. Choose the truncation order asymptotically: terms of a divergent Bernoulli series should be stopped near their least magnitude, not at a fixed arbitrarily high order.

Chapter 18

Conjectures and research directions

The proved formulae expose several extensions that appear both tractable and mathematically substantive. Conjectures below are deliberately separated from theorems.

18.1 A uniform coalescing root-of-unity normal form

The fixed-a theorem (12.3) fails uniformly as $a^m \to 1$, while shifted root-of-unity formulae reorganize the same object in a scale containing $\log t$ and gamma factors. The recent exact gamma-product formula of Arabi Ardehali–Rosengren provides a new route to such uniformity near $q = 1$ [6].

Conjecture 18.1 (Uniform cyclic-gamma normal form)**.** *Let ζ be primitive of order m, let $q = \zeta e^{-t}$, and let $a = a(t)$ satisfy $1 - a(t)^m = \mathcal{O}(t^\sigma)$ for a fixed $\sigma > 0$. After extracting an explicit product of*

(i) $\exp(-\operatorname{Li}_2(a^m)/(m^2 t))$,

(ii) *a cyclic quantum dilogarithm,*

(iii) *finitely many ordinary gamma factors in the scaled variable $(1 - a^m)/t$, and*

(iv) *an elementary power of t,*

there is a single coefficient expansion, uniform for the scaled variable in compact subsets of a cut plane, that reduces to (12.3) when $(1 - a^m)/t \to \infty$ and to the shifted root-of-unity expansion when $(1 - a^m)/t = \mathcal{O}(1)$.

A proof should follow from residue-class dissection plus an exact Euler–Maclaurin formula with a singular endpoint retained rather than expanded. This would eliminate the current need to choose between "fixed parameter" and "coalescing parameter" cases.

18.2 Higher q-Lucas jets

The value at a root of unity is controlled by q-Lucas, and the zero order is a single carry. The theorem at $q = -1$ shows that the first derivative also factorizes into a small polynomial in quotient digits times an ordinary binomial coefficient.

Conjecture 18.2 (Polynomial cyclotomic jets)**.** *Fix m, a primitive mth root ζ, residues $0 \le r, s < m$, and a jet order J. Write $n = am + r$ and $k = bm + s$. After extracting the possible simple factor $q - \zeta$ and the natural leading q-Lucas factor, the first J normalized Taylor coefficients of $\begin{bmatrix} n \\ k \end{bmatrix}_q$ at $q = \zeta$ are polynomials in a and b with coefficients in $\mathbb{Q}(\zeta)$, of degree bounded linearly in J.*

Block the factorial products into intervals of length m. Each block has a Taylor logarithm polynomial in the block index, so Faulhaber summation strongly suggests the conjecture. A proof would give a genuine "differential q-Lucas theorem" and an efficient algorithm independent of the size of n.

18.3 Rigorous complex double scaling

Theorem 15.2 is proved for positive real τ with uniformity on compact sets. The same Euler–Maclaurin derivation should extend to complex τ wherever the segments $\{\beta\tau : 0 \le \beta \le 1\}$ arising in the endpoint integrals avoid the nonzero lattice $2\pi\mathrm{i}\mathbb{Z}$. Natural limiting cuts are therefore the imaginary rays beginning at $\pm 2\pi\mathrm{i}$ (or at $\pm 2\pi\mathrm{i}/\beta$ for an individual endpoint parameter β). For finite N, zeros and poles occur at rational points corresponding to roots of unity; as N grows, these points become dense along the limiting cuts. A worthwhile next step is a uniform theorem on compact subsets of the cut τ-plane, followed by matched asymptotics as a cut is approached.

The double-scaling action $\Phi(\alpha, \tau)$ is also a natural candidate for a large-deviation rate function for inversion statistics under exponential tilting. Differentiation with respect to $\log q = -\tau/N$ gives limiting cumulant densities, while Legendre transformation should produce an explicit dilogarithmic rate function.

18.4 Resurgence and exponentially small sectors

Bernoulli coefficients grow factorially, so the fixed-a asymptotic series is normally divergent after analytic continuation beyond its local convergence setting. Exact eta modularity identifies the complete exponentially small sector in one special case. Recent work shows that suitable q-Pochhammer combinations possess resurgent and quantum-modular structures [7, 8].

Promising concrete tasks are:

(a) compute the Borel transform of the coefficient series in (12.3) for general m and locate its singularities;

(b) express the Stokes constants directly through residue-class Fourier transforms of $\mathcal{E}_j(\zeta^r)$;

(c) determine when Dirichlet-character weighted cyclic products cancel all but one tower of Borel singularities;

(d) compare the exact gamma-product completion [6] with median Borel summation.

18.5 Automatic proof-producing expansion

The finite theory is particularly suitable for Lean formalization. One can build a verified expansion compiler with the following pipeline:

$$\text{product or cyclotomic factorization} \longrightarrow \text{logarithmic jets} \longrightarrow \text{Bell recurrence} \longrightarrow \text{Taylor certificate.}$$

For Gaussian coefficients at $q = 1$, Bernoulli power sums replace polynomial factorization; at roots of unity, exact arithmetic takes place in a cyclotomic extension. The certificate can be an identity in a truncated power-series ring, so no analytic infrastructure is needed for finite products. Analytic remainder bounds can then be layered on separately.

18.6 Optimal truncation and certified error bounds

For numerical work, an asymptotic formula is most valuable when paired with a computable remainder estimate. The fixed-a Lambert proof already gives a simple geometric bound, but it is far from sharp at high order. A useful research target is an adaptive bound that combines:

(i) the distance to the nearest singularity in the t-plane;

(ii) exact Bernoulli-number bounds;

(iii) the magnitude of the relevant polylogarithm;

(iv) an exponentially small tail estimated through a modular or gamma-product transform.

Such a bound would automatically select the least term and produce a certified number of correct digits.

Chapter 19

Conclusions

The expansions of q-analogues are governed by a small number of reusable mechanisms.

First, finite products are algebraic. Logarithmic derivatives plus Bell polynomials solve every regular local problem, while factor extraction solves every resonant one. At $q = 0$, the same products become weighted partition generators; at infinity, reciprocity reverses those coefficients. Cyclotomic factorization is the correct coordinate system at roots of unity.

Second, the coordinate $q = \mathrm{e}^{-t}$ linearizes the multiplicative geometry near $q = 1$. Bernoulli numbers enter through the universal factor $1 - \mathrm{e}^{-x}$, power sums record finite endpoints, and polylogarithms record infinite Lambert sums. Bell polynomials then convert logarithmic data into ordinary coefficients. This single scheme covers finite shifted factorials, Gaussian and multinomial coefficients, q-gamma and q-beta functions, and basic hypergeometric series.

Third, modularity and roots of unity require information beyond algebraic powers. Eta transformations expose exact exponentially small corrections, while residue-class splitting produces cyclic quantum dilogarithms. The boundary between fixed and coalescing parameters is the main remaining uniformity problem.

Finally, the scale $q = \mathrm{e}^{-\tau/N}$ shows how fixed-degree perturbation theory merges with large combinatorial systems. Its all-order odd inverse-power expansion is both a q-analogue of Stirling's formula and a dilogarithmic large-deviation expansion. Together with the exact cyclotomic jets, it offers a concrete path from symbolic identities to certified asymptotic computation and formal proof.

Appendix A

Coefficient extraction and conversion tables

A.1 From a logarithmic expansion to an ordinary expansion

The most economical way to calculate high-order coefficients is to retain the logarithm until the final step. The following recurrence is equivalent to the complete exponential Bell-polynomial formula, but it is usually faster in a computer algebra system and simpler to formalize.

Proposition A.1 (Logarithm-to-product recurrence)**.** *Suppose*

$$L(z) = \sum_{r\geq 1} \ell_r z^r, \qquad \exp L(z) = \sum_{m\geq 0} c_m z^m.$$

Then $c_0 = 1$ and, for $m \geq 1$,

$$c_m = \frac{1}{m} \sum_{r=1}^{m} r \ell_r c_{m-r}. \tag{A.1}$$

If instead $L(z) = \sum_{r\geq 1} \kappa_r z^r / r!$, then

$$c_m = \frac{1}{m!} \mathcal{Y}_m(\kappa_1, \ldots, \kappa_m),$$

where $\mathcal{Y}_m$ is the complete exponential Bell polynomial.

Proof. Differentiate $C(z) = \exp L(z)$ to obtain $C' = L'C$. Equating the coefficient of z^{m-1} gives $mc_m = \sum_{r=1}^{m} r\ell_r c_{m-r}$. The Bell-polynomial statement is the standard exponential generating-function form of the same identity. □

When the original function has a zero of order ν, first write

$$F(z) = z^\nu C \exp L(z), \qquad C \neq 0,$$

and apply the recurrence to the unit factor. This one preliminary extraction is what makes the same algorithm valid at regular and resonant points.

A.2 Conversion between $t = -\log q$ and $h = q - 1$

Let $s(m, r)$ denote the signed Stirling number of the first kind, determined by

$$x(x-1)\cdots(x-m+1) = \sum_{r=0}^{m} s(m,r)x^r.$$

If

$$F(\mathrm{e}^{-t}) = \sum_{r\geq 0} a_r \frac{t^r}{r!}, \qquad q = 1+h,$$

then

$$F(1+h) = \sum_{m\geq 0} \left(\sum_{r=0}^{m} (-1)^r s(m,r) a_r \right) \frac{h^m}{m!}. \tag{A.2}$$

Indeed,

$$\frac{(-\log(1+h))^r}{r!} = (-1)^r \sum_{m\geq r} s(m,r) \frac{h^m}{m!}.$$

Thus a complete expansion in the geometrically natural variable t can be translated exactly to a literal Taylor expansion in $q-1$ by a finite triangular transform at every order.

A.3 Gaussian quick-reference coefficients

Let

$$D = k(n-k), \qquad \Delta_{2r} = S_{2r}(n) - S_{2r}(k) - S_{2r}(n-k).$$

The first three Faulhaber differences are

$$\begin{aligned} \Delta_2 &= D(n+1), \\ \Delta_4 &= D(n+1)\big(n(n+1) - D\big), \\ \Delta_6 &= \frac{D(n+1)}{2}\big(2D^2 - 4Dn^2 - 5Dn + 2n^4 + 4n^3 + n^2 - n\big). \end{aligned}$$

Consequently, for the centered normalized coefficient

$$H_{n,k}(t) = \mathrm{e}^{Dt/2} \frac{\left[n \atop k \right]_{\mathrm{e}^{-t}}}{\binom{n}{k}},$$

we have

$$\begin{aligned} \log H_{n,k}(t) &= \frac{\Delta_2}{24} t^2 - \frac{\Delta_4}{2880} t^4 + \frac{\Delta_6}{181440} t^6 + \mathcal{O}(t^8), \\ H_{n,k}(t) &= 1 + A_2 t^2 + \big(A_4 + \tfrac{1}{2} A_2^2\big) t^4 \\ &\qquad + \big(A_6 + A_2 A_4 + \tfrac{1}{6} A_2^3\big) t^6 + \mathcal{O}(t^8), \end{aligned}$$

where

$$A_2 = \frac{\Delta_2}{24}, \qquad A_4 = -\frac{\Delta_4}{2880}, \qquad A_6 = \frac{\Delta_6}{181440}.$$

In the literal coordinate $h = q - 1$, the uncentered normalized polynomial has

$$\frac{\left[n \atop k \right]_{1+h}}{\binom{n}{k}} = 1 + \frac{D}{2} h + \frac{D(3D+n-5)}{24} h^2 + \frac{D(D-2)(D+n-3)}{48} h^3$$

$$
\begin{aligned}
&+ \frac{D}{5760}\Big(15D^3 + 30D^2 n - 150D^2 + 5Dn^2 - 168Dn + 487D \\
&\qquad - 2n^3 - 4n^2 + 218n - 500\Big)h^4 + \mathcal{O}(h^5). \qquad \text{(A.3)}
\end{aligned}
$$

Formula (A.3) follows mechanically from (4.1), (A.2), and (A.1); it is included as a useful regression test for symbolic implementations.

Appendix B

Small-base and reciprocal coefficient tables

B.1 A stable table for $(aq;q)_\infty$

The weighted distinct-partition expansion gives

$$\begin{aligned}(aq;q)_\infty = 1 - aq - aq^2 + a(a-1)q^3 + a(a-1)q^4 \\ + a(2a-1)q^5 - a(a-1)^2q^6 - a(a^2-3a+1)q^7 \\ - a(a-1)(2a-1)q^8 + \mathcal{O}(q^9).\end{aligned}$$

Multiplication by $1-a$ gives the corresponding expansion of $(a;q)_\infty$. The coefficient of q^r is already stable in the finite product $(aq;q)_{n-1}$ when $n > r$.

At $a = 1$, Euler's pentagonal cancellation turns the table into

$$(q;q)_\infty = 1 - q - q^2 + q^5 + q^7 + \mathcal{O}(q^{12}).$$

At $a = -1$, it gives the generating product for partitions into distinct parts,

$$(-q;q)_\infty = 1 + q + q^2 + 2q^3 + 2q^4 + 3q^5 + 4q^6 + 5q^7 + 6q^8 + \cdots.$$

B.2 Reciprocal transfer

If a finite polynomial $P(q) = \sum_{r=0}^{D} c_r q^r$ is reciprocal, $q^D P(q^{-1}) = P(q)$, then the same coefficient list describes both ends:

$$P(q) = c_0 + c_1 q + \cdots + c_D q^D = q^D \left(c_0 + c_1 q^{-1} + \cdots + c_D q^{-D}\right).$$

For Gaussian coefficients $c_r = p_{k,n-k}(r)$. Hence every finite small-q coefficient theorem immediately becomes a large-q theorem, with the parity factor $(-1)^D$ inserted on the negative real ray.

Appendix C

Reproducible symbolic and numerical checks

C.1 Companion program

The archive contains `q_expansion_experiments.py`. It requires only Python, `sympy`, and `mpmath`; no specialized q-series package is used. Run it from the archive directory with

```
python q_expansion_experiments.py
```

The program performs the following independent checks.

(i) It constructs $\begin{bmatrix} n \\ k \end{bmatrix}_q$ over $\mathbb{Z}[q]$ by the q-Pascal recurrence, checks the normalized expansion at $q = 1$ through fourth order, and checks the value and derivative formula at $q = -1$, in each case for all $0 \le k \le n \le 20$.

(ii) It derives Δ_2, Δ_4, and Δ_6 symbolically from Faulhaber sums and reduces them to polynomials in n and $D = k(n-k)$.

(iii) It expands $\prod_{j \ge 1}(1 - aq^j)$ through q^8 by exact polynomial arithmetic.

(iv) It compares the Mellin expansion of $(q^x; q)_\infty$, the q-gamma expansion, the radial root-of-unity formula, and the double-scaling Gaussian formula with direct high-precision evaluation.

C.2 Recorded test instance

With 80-decimal-digit working precision, the included run produced

Test	Parameters and truncation	Absolute or signed error
$(q^x; q)_\infty$ logarithm	$x = 2.5$, $t = 0.1$, through t^6	$2.64769017918096558 \times 10^{-14}$
$\log \Gamma_q(x)$	$x = 3.7$, $t = 0.08$, through t^6	$5.00835558895922029 \times 10^{-13}$
Primitive cubic root	$a = 0.2$, $t = 0.05$, through t^6	$3.4805992014908836 \times 10^{-11}$
Double scaling	$N = 100$, $\alpha = 0.4$, $\tau = 1.7$, through N^{-1}	$5.34631320464741412 \times 10^{-8}$
Double scaling	same parameters, through N^{-3}	$-8.68801964159948774 \times 10^{-12}$

Each exact $q = 1$ and $q = -1$ suite covered 230 index pairs and found no discrepancy. These checks do not replace the proofs, but they are sensitive to signs, Bernoulli normalizations, branch choices, and endpoint corrections that are easy to get wrong in hand calculations.

Appendix D

Compact formula map

For reference, the principal expansion engines can be summarized in one page.

Regime	Natural normalization	Coefficient engine
Finite, regular $q = q_0$	Divide by the nonzero value at q_0	Logarithmic derivatives and Bell recurrence
Finite, resonant $q = q_0$	Extract $(q - q_0)^\nu$	Nonvanishing-factor logarithmic derivatives
$q \to 1$, fixed degree	$q = \mathrm{e}^{-t}$; divide by the classical limit	Bernoulli numbers and Faulhaber differences
$q \to 0$	None	Weighted restricted partitions
$q \to \infty$	Divide by the leading monomial	Reciprocity and the small-q table
Finite root of unity	Extract the cyclotomic zero	Cyclotomic exponents, residue classes, Bell recurrence
Infinite $q \to 1$, fixed a	Remove $\exp(-\operatorname{Li}_2(a)/t)$	Lambert series and polylogarithms
Infinite $q \to 1$, $a = q^x$	Remove exponential, power, and gamma factors	Mellin residues and Bernoulli polynomials
Radial root of unity	Remove $\exp(-\operatorname{Li}_2(a^m)/(m^2 t))$	Resonant residue class plus cyclic dilogarithm
$q = \mathrm{e}^{-\tau/N}$	Remove the order-N dilogarithmic action	Singular Euler–Maclaurin; odd powers of N^{-1}

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